

Article

Maximizing Return on Investment in Cryptocurrency Mining Through Energy Optimization

Mohammad Nasrinasrabadi ¹, Maryam A. Hejazi ^{1,*}, Arefeh Jaberri ¹, Hamed Hashemi-Dezaki ¹
and Hossein Shahinzadeh ^{2,*}

¹ Department of Electrical and Computer Engineering, University of Kashan, Kashan 8731753153, Iran; mohammad.nasrinasrabadi@gmail.com (M.N.); a.jaberri6574@gmail.com (A.J.); hamed.hashemi@kashanu.ac.ir (H.H.-D.)

² Department of Electrical Engineering, Amirkabir University of Technology (Tehran Polytechnic), Tehran 3715879817, Iran

* Correspondence: mhejazi@kashanu.ac.ir (M.A.H.); h.s.shahinzadeh@ieee.org (H.S.)

Abstract

Cryptocurrencies utilize blockchain technology to ensure transparency, decentralization, and immutability in financial transactions. It is expected that blockchain applications will significantly impact renewable energy markets. However, there is a lack of studies addressing the energy requirements of digital currencies. This research proposes optimizing a hybrid energy system consisting of distributed renewable and non-renewable energy sources, focusing on cryptocurrency mining. Although previous studies have not yet addressed energy system optimization considering cryptocurrency mining farms, the increasing prominence of such farms highlights the growing need for research in this area. The primary renewable sources in the proposed hybrid system include photovoltaic (PV) panels and wind turbines. We employ diesel generators as backup systems to compensate for the intermittent nature of solar and wind energy production. Besides meeting the demands of urban loads, cryptocurrency mining devices will be considered a major energy consumer. In this article, the optimal configuration of the energy system will be determined based on technical and economic indicators. Additionally, economic evaluations will be conducted to assess the income generated from cryptocurrency mining farms, and appropriate approaches will be identified from both technical and financial perspectives, focusing on return on investment (ROI).

Keywords: hybrid energy systems; renewable energy; distributed generation sources; cryptocurrency mining farms; energy system optimization



Academic Editor: Tek-Tjing Lie

Received: 9 July 2025

Revised: 23 September 2025

Accepted: 25 September 2025

Published: 10 November 2025

Citation: Nasrinasrabadi, M.; A. Hejazi, M.; Jaberri, A.; Hashemi-Dezaki, H.; Shahinzadeh, H. Maximizing Return on Investment in Cryptocurrency Mining Through Energy Optimization. *Energies* **2025**, *18*, 5910. <https://doi.org/10.3390/en18225910>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

1.1. Background

The growing demand for renewable energy and the rise in cryptocurrency mining have highlighted the need for sustainable energy solutions. Renewable sources like solar and wind offer environmental benefits but face challenges due to their intermittent nature. Cryptocurrency mining, known for its high energy consumption, presents ecological concerns, especially when powered by fossil fuels. Integrating renewable energy with mining operations offers a viable solution, reducing costs, environmental impact, and enhancing profitability. This synergy enables mining operations to optimize energy usage while supporting global efforts toward a sustainable and low-carbon energy future.

1.2. Problem Statement

Cryptocurrency mining is notorious for its immense energy consumption, primarily due to the Proof-of-Work (PoW) consensus mechanism. As the industry grows, this energy demand strains power grids and exacerbates environmental concerns, particularly in regions reliant on fossil fuels. Although renewable energy sources offer a more sustainable alternative, their intermittent nature makes a consistent power supply challenging, especially for energy-intensive operations like cryptocurrency mining.

Despite the potential for renewable energy to power mining farms, limited research addresses the optimization of hybrid energy systems specifically designed for cryptocurrency mining. The objective of this research is to develop an optimized hybrid energy system that integrates renewable (solar and wind) and non-renewable (diesel) energy sources to reduce costs and maximize ROI for cryptocurrency mining operations. This study fills the gap by proposing a solution that balances energy supply and demand while minimizing environmental impact and ensuring the economic feasibility of mining operations.

In this study, other fossil energy sources such as natural gas and coal were not included due to local and technical restrictions. In Iran, using natural gas for small-scale systems like cryptocurrency mining requires strict government permits, and frequent shortages limit its use for such applications. Coal is also not a practical option because of its high pollution, low availability, and mismatch with sustainable energy goals. From a technical point of view, distributed generators with a capacity of up to one megawatt mostly operate on diesel fuel, which has reliable infrastructure and is one of the cheapest fuels available in Iran. For these reasons, this research focuses only on PV panels, wind turbines, and diesel generators as the most practical and cost-effective solution for cryptocurrency mining energy systems.

1.3. Contributions

This paper offers several distinct contributions to optimizing energy systems for cryptocurrency mining. First, it proposes a novel hybrid energy model that integrates PV panels, wind turbines, and diesel generators to meet the continuous and high-energy demands of cryptocurrency mining operations. Second, it provides an in-depth economic analysis that evaluates the ROI from mining farms, emphasizing how the strategic use of renewable energy reduces operational costs and dependence on fossil fuels. Third, the study addresses a critical research gap by optimizing the configuration of hybrid energy systems, considering both technical efficiency and economic performance, which has yet to be thoroughly explored in existing research. Finally, the work demonstrates the potential for large-scale adoption of renewable energy in cryptocurrency mining, contributing to reducing carbon emissions while maintaining profitability.

To summarize, the key contributions of this study are as follows:

- **Novel Hybrid Energy Model:** The study proposes a hybrid energy system that integrates PV panels, wind turbines, and diesel generators to meet the continuous and high-energy demands of cryptocurrency mining operations.
- **Economic Analysis of ROI:** It provides a detailed economic analysis, evaluating the ROI for mining farms, and highlights how renewable energy can reduce operational costs and reliance on fossil fuels.
- **Optimization of Hybrid Systems:** The research addresses a gap in existing literature by optimizing the configuration of hybrid energy systems, balancing technical efficiency and economic performance specifically for cryptocurrency mining.
- **Large-Scale Renewable Energy Adoption:** The article demonstrates the feasibility and potential of adopting renewable energy on a large scale in cryptocurrency mining, achieving reduced carbon emissions while maintaining profitability.

1.4. Structure of the Paper

The rest of the paper is organized as follows: Section 2 provides a literature review on distributed energy resources (DERs) integration and energy systems in the context of cryptocurrency mining. Section 3 describes the system modeling and methodology for the proposed hybrid energy system. Section 4 presents the proposed optimization algorithm, detailing its design and innovations. Section 5 discusses a case study, including system description, scenario analysis, and simulation results. Section 6 outlines the discussion, covering the interpretation of results and limitations of the study. Section 7 offers recommendations for future research. Section 7 concludes the paper. The structure of the paper is illustrated in Figure 1. Table 1 clearly compares our research and existing related review papers.

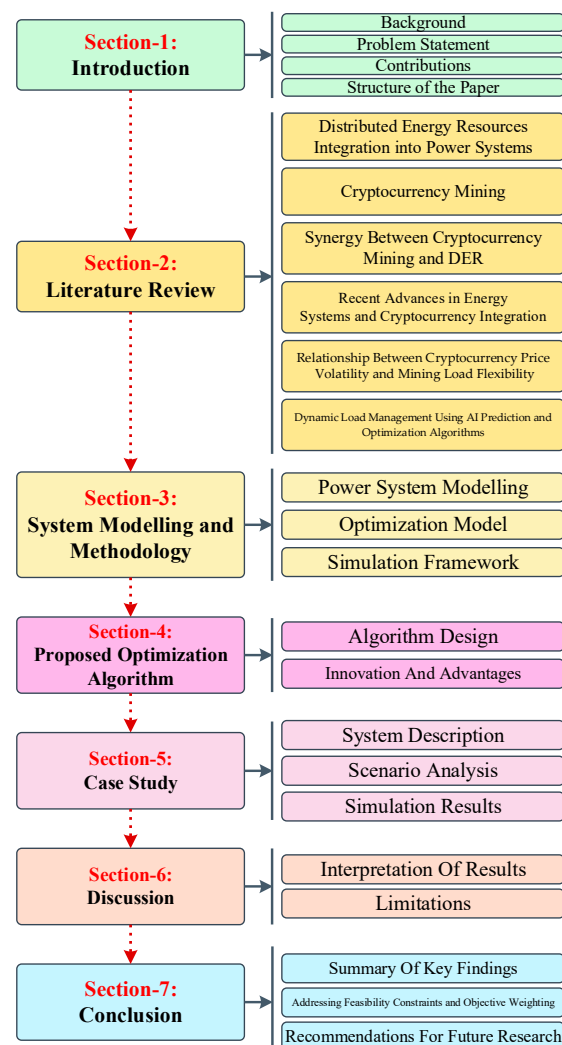


Figure 1. The structure of this paper.

Table 1. Literature Review of Energy Systems in the Presence of Cryptocurrency.

Reference	Year of Publication	PV	Wind Turbine	Diesel Generator	Storage	Converter	Load Profile	HOMER Pro Software 3.14.7524.23887	Cryptocurrency
Bekele & Boneya [1]	2012	Yes	Yes	No	No	No	No	Yes	No
Bentouba & Bourouis [2]	2016	Yes	Yes	Yes	No	No	No	No	No
Ajlan et al. [3]	2017	Yes	Yes	No	Yes	No	Yes	Yes	No
Rajbongshi et al. [4]	2017	Yes	No	Yes	No	No	Yes	No	No
Hossain et al. [5]	2017	Yes	Yes	Yes	Yes	Yes	No	Yes	No
de Vries [6]	2018	No	No	No	No	No	No	No	Yes
Humbatova et al. [7]	2019	No	No	No	No	No	No	No	Yes
Krishan & Suhag [8]	2019	Yes	Yes	No	Yes	No	No	Yes	No
Stoll et al. [9]	2019	No	No	No	No	No	No	No	Yes
Kristoufek [10]	2020	No	No	No	No	No	No	No	Yes
Das & Dutta [11]	2020	No	No	No	No	No	No	No	Yes
Gallersdörfer et al. [12]	2020	Yes	Yes	No	No	No	No	No	Yes
Karmakar et al. [13]	2021	No	No	No	No	No	Yes	No	Yes
Hojjat & Kalali [14]	2021	Yes	No	No	No	No	No	No	Yes
Khalil et al. [15]	2021	Yes	Yes	No	No	Yes	Yes	Yes	No
Náñez Alonso et al. [16]	2021	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Nikzad & Mehregan [17]	2022	Yes	No	No	No	Yes	Yes	No	Yes
Menati et al. [18]	2023	No	No	No	No	No	Yes	No	Yes
Chamanara et al. [19]	2023	No	No	No	No	No	No	No	Yes
A. Rudd et al. [20]	2023	No	No	No	No	No	No	No	Yes
Hajiaghapour-Moghim et al. [21]	2024	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Hakimi et al. [22]	2024	Yes	No	No	No	Yes	Yes	No	Yes
Laimon et al. [23]	2025	No	No	No	No	No	No	No	Yes
B. Ding et al. [24]	2025	No	Yes	No	Yes	No	Yes	No	No
Our Work	2025	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

2. Literature Review

2.1. Distributed Energy Resources Integration into Power Systems

The integration of DERs into power systems poses both technical and economic challenges. Recent studies have examined various aspects of DER integration, such as voltage control, microgrids, and the interaction of distributed generation with centralized systems. For instance, advanced voltage regulation techniques in DER-based microgrids are crucial for maintaining system stability and mitigating issues like voltage sags and harmonic distortions [25].

Key challenges include the intermittent nature of renewable energy sources such as wind and solar, which cause variability in power production. This often leads to overloads, voltage imbalances, and reverse power flows in grids with high DER penetration [26].

Technologies such as advanced control systems and Distributed Energy Resource Management Systems (DERMS) have been developed to facilitate better grid stability and efficiency during DER integration. These systems help manage grid operations, enhancing both reliability and economic performance [27].

Furthermore, operational challenges like balancing supply and demand amid unpredictable renewable energy outputs demand innovative planning strategies. Real-time data utilization and multi-objective optimization techniques have proven essential in improving grid resilience and operational efficiency [28].

In conclusion, the integration of DERs requires continuous research, focusing on both technical enhancements and economic feasibility to address the complexities of modern power systems.

2.2. Cryptocurrency Mining

Cryptocurrency mining, particularly Bitcoin mining, is recognized for its significant energy consumption, with annual energy usage exceeding that of entire nations like Portugal. The energy-intensive nature of this process is mainly driven by the PoW consensus

algorithm, which demands immense computational power, leading to high electricity consumption and substantial carbon emissions [29]. In contrast, alternative consensus mechanisms like Proof-of-Stake (PoS) significantly reduce energy usage by removing the need for continuous computational competition. Recent sustainability strategies in mining also include energy-efficient cooling systems, the use of surplus renewable energy, and AI-based optimization of mining operations to minimize environmental harm [30].

Recent studies reveal that Bitcoin mining alone is responsible for emitting up to 90.2 million tons of CO₂ annually, positioning it as a serious environmental threat. The energy consumption of Bitcoin and other cryptocurrencies is heavily influenced by the volatility in cryptocurrency prices, such as Bitcoin and Ethereum, which directly affects the electricity used for mining [31].

The environmental impact of cryptocurrency mining varies significantly depending on the location of mining operations. In regions heavily dependent on fossil fuels, such as China, cryptocurrency mining contributes considerably to carbon emissions. In contrast, regions that utilize renewable energy sources show a reduced environmental footprint [32].

There is a growing focus on sustainable cryptocurrency mining practices. For example, studies indicate that utilizing hydropower for Bitcoin mining can significantly decrease its carbon footprint, making it a more environmentally friendly alternative to fossil fuel-powered mining [33].

In summary, while cryptocurrency mining presents substantial challenges in terms of energy consumption and environmental impact, the transition towards renewable energy in mining operations offers a promising avenue for mitigating these effects.

2.3. Synergy Between Cryptocurrency Mining and DER

The integration of DERs into power systems presents significant challenges and opportunities. On one hand, cryptocurrency mining, especially Bitcoin mining, demands high energy consumption, straining local power grids. However, the synergy between cryptocurrency mining and DERs, such as solar and wind energy, can optimize energy efficiency and reduce environmental impact [34].

2.3.1. Opportunities

- **Renewable Energy Usage:** Integrating renewable energy sources like PV panels and wind for cryptocurrency mining has the potential to reduce environmental footprints. Research shows that using distributed renewable energy for mining operations increases ROI and reduces emissions [32]. Additionally, innovative strategies aimed at reducing the environmental impact of cryptocurrency mining include the adoption of energy-efficient consensus mechanisms and the utilization of surplus renewable energy that would otherwise be curtailed due to grid limitations. Laimon and Almadadha [23] advocate for the use of the Stellar Consensus Protocol in combination with system dynamics modeling to enhance the efficiency of renewable energy integration in crypto mining operations. Their approach outlines sustainable pathways to reduce carbon emissions and improve the overall environmental sustainability of blockchain systems.
- **Combined Heat and Power (CHP) Systems:** Mining operations can utilize CHP systems, which generate electricity while capturing heat for other uses, thus improving overall energy efficiency. This approach can reduce the energy required for cooling mining rigs and enhance the performance of cryptocurrency mining farms [35].
- **Demand Flexibility:** Cryptocurrency mining facilities can offer demand-side flexibility to balance electricity grids. By adjusting mining loads in response to grid needs, these

operations can help absorb excess renewable energy during off-peak times or reduce consumption during peak demand [36].

2.3.2. Challenges

- **Energy Consumption:** A significant challenge is the sheer amount of energy required for mining. PoW cryptocurrencies like Bitcoin are particularly demanding, contributing to high electricity consumption and carbon emissions. This raises concerns about grid stability, especially in regions with less robust infrastructure [37].
- **Regulatory and Pricing Issues:** Pricing strategies, such as time-of-use (TOU) or increased electricity rates during peak hours, may help limit the profitability of home-based miners. These measures could manage demand surges caused by residential mining, though implementation remains a challenge in some regions [38].
- **Grid Stability and Management:** The integration of DERs with cryptocurrency mining can lead to voltage imbalances, overloads, and reverse power flows in the system. To mitigate this, innovative energy management systems and demand response strategies are needed [39].

2.4. Recent Advances in Energy Systems and Cryptocurrency Integration (2022–2025)

The growing convergence of cryptocurrency technologies and modern energy systems has prompted a wave of academic interest focused on understanding the environmental, technical, and regulatory implications of blockchain-based energy consumption. This section synthesizes recent peer-reviewed research (2022–2025) to highlight key trends, challenges, and advancements in the integration of cryptocurrency with renewable energy and smart grid systems.

2.4.1. Cryptocurrency Mining and Environmental Impact

Recent studies reveal the outsized environmental footprint of cryptocurrency mining, especially from PoW algorithms such as Bitcoin. According to a comprehensive analysis, cryptocurrency mining currently accounts for approximately 0.5% of global electricity consumption and 0.25% of global CO₂ emissions, with projections indicating a potential sixfold increase in energy usage by 2030 if mitigation strategies are not adopted. The carbon intensity of a single Bitcoin transaction can reach 712 kg CO₂, comparable to driving over 2000 km in a gasoline vehicle [23].

In addition to CO₂ emissions, mining operations exert considerable pressure on water resources, with studies showing a 53.7% increase in Bitcoin's water footprint between 2020 and 2021 [40]. These findings underscore the need for more energy and resource-efficient blockchain protocols.

2.4.2. Renewable Energy Integration in Mining

Efforts to mitigate the environmental burden of mining have led to a growing body of work on integrating renewable energy into mining operations. Research on hybrid systems, such as combining PV panels and hydropower with mining farms, shows promising results in lowering operational costs and reducing emissions [41]. A systems dynamics model applied to grid scenarios confirms that renewable-powered mining can contribute to grid balancing and curtailment reduction, especially during off-peak hours [42].

2.4.3. Evolution of Consensus Mechanisms

An essential thread in the literature is the evolution from PoW to PoS and other low-energy consensus mechanisms. Ethereum's transition to PoS, for instance, resulted in a 99.95% reduction in energy usage, demonstrating the viability of alternative models for sustainable blockchain operation [43,44]. Comparative analyses further suggest that PoS

and Proof-of-Space-Time protocols are capable of maintaining network security with only a fraction of the energy demand [45].

2.4.4. Blockchain and IoT in Energy Systems

Blockchain's synergy with IoT has transformed real-time monitoring, peer-to-peer (P2P) trading, and decentralized energy resource management. Studies show that blockchain-based P2P energy platforms can support over 90% of energy trades using smart contracts while improving transparency and reducing transaction costs [46,47]. In smart grid applications, blockchain-enabled IoT platforms demonstrated the ability to manage up to 40% more variability in renewable energy production compared to conventional systems [48].

These findings are consistent with broader reviews that emphasize the potential of blockchain-IoT integration to enhance grid scalability, security, and automation [44].

In summary, the recent literature underscores a decisive shift toward environmentally conscious cryptocurrency systems, driven by the adoption of energy-efficient consensus mechanisms and renewable energy integration, which collectively enhance grid sustainability, reduce operational emissions, and enable smarter, decentralized energy management.

2.5. Relationship Between Cryptocurrency Price Volatility and Mining Load Flexibility

Recent scholarship has increasingly examined how fluctuations in major cryptocurrency prices—principally Bitcoin and Ethereum—shape miners' load-flexibility behaviors and, in turn, influence grid operations and energy markets. Sapra and Shaikh (2023) pioneered econometric modeling of the interplay between Bitcoin mining load and market determinants, finding that electricity-price and gas-price volatility both materially drive mining energy consumption, but their framework did not fully capture miners' temporal responsiveness to short-lived price swings [49]. Building on this, Karmakar et al. (2021) employ a GARCH-X model to demonstrate that periods of heightened Bitcoin price volatility coincide with measurable ramp-down events in large mining facilities, indicating an economic load response where miners curtail operations to preserve profitability under adverse price shocks [13,50].

Similarly, Menati et al. (2022) simulate Texas ERCOT grid conditions with integrated synthetic mining loads and quantify the profit-maximizing deployment of flexible mining capacity under varying Bitcoin price and electricity-cost scenarios; their findings reveal temporal mining elasticity—miners shift consumption to low-price hours and offer ancillary services during peak prices, delivering up to 8 \$/MWh of demand response revenues per MW of flex capacity [36].

On a system level, Bouri et al. (2021) [51] link high crypto volatility episodes to amplified net demand variability in interconnected grids, exacerbating price spikes and raising reserve-margin requirements by an estimated 5% during intense speculative cycles. Their vector-autoregression analysis further shows that volatility spillovers from crypto to energy markets increase net peak load deviations by 10–15% in regions with substantial mining presence [51,52]. However, these studies often assume homogeneous miner behavior and static flexibility parameters.

To address these limitations, Lu et al. (2023) [53] develop an agent-based model of heterogeneous mining operators with distinct risk tolerances and dynamic operational constraints; they demonstrate that allowing miners to adjust load in real time based on live price signals reduces system-wide price volatility by 12% and decreases unserved energy events by 3% relative to fixed-load baselines. Likewise, recent work by SSRN (2025) formalizes a bi-level optimization capturing both miner profit-maximization and grid reliability objectives, showing that coordinated flexibility markets can yield up to

20% additional downward load shifting during price spikes without compromising miner revenue [53,54].

Collectively, these advances move beyond Sapra and Shaikh's initial econometric approach by incorporating stochastic volatility modeling, high-resolution temporal elasticity, and system-level integration of flexible mining loads, underscoring the vital role of dynamic economic load response in aligning mining operations with grid stability goals.

2.6. Dynamic Load Management Using AI Prediction and Optimization Algorithms

Dynamic load management in energy systems with integrated cryptocurrency mining has witnessed rapid evolution through AI-based prediction and optimization techniques. Deep learning methods, particularly LSTM (Long Short-Term Memory) and CNN (Convolutional Neural Network) architectures, have been applied to forecast mine-specific load profiles and renewable generation patterns. For instance, El-Hendawi and Wang (2020) utilize LSTM networks for short-term load prediction in microgrids with flexible demands, achieving 8% lower RMSE (Root Mean Square Error) compared to baseline ARIMA (AutoRegressive Integrated Moving Average) models [55].

In reinforcement-learning frameworks, Nakabi and Toivanen (2021) compare DQN, DDGP, and PPO agents for real-time energy management in microgrids with flexible loads; their simulations demonstrate that PPO yields the best trade-off between cost minimization and constraint adherence, reducing peak-to-average load ratios by 15% [56]. Hybrid approaches combining model predictive control (MPC) with RL have also emerged: Chen et al. (2020) propose an MPC-guided DDGP agent that respects voltage and frequency constraints while optimizing load scheduling, improving renewable utilization by 10% over pure MPC [56,57].

Recent studies in IEEE Transactions on Smart Grid highlight multi-agent RL for coordinated dispatch of mining loads alongside residential and industrial consumers. A 2024 paper employs a federated Q-learning scheme enabling distributed miners to learn price-responsive policies without sharing proprietary data, which results in a 20% reduction in grid imbalance penalties while preserving miner privacy.

Hybrid machine-learning and evolutionary optimization frameworks have further enhanced dynamic load management. A 2025 Applied Energy article integrates CNN-based price predictors with genetic algorithms for scheduling mining operations, yielding 5% additional profit under volatile price conditions and 7% lower load variability [58]. Additionally, multi-objective optimization using particle-swarm and moth-flame algorithms has been applied to islanded microgrids with mining, demonstrating resilience improvements under renewable intermittency by simultaneously minimizing curtailment, emissions, and cost [59].

Collectively, these AI-driven advances—spanning deep learning forecasting, reinforcement and hybrid optimization algorithms—substantially enhance the dynamic management of price-sensitive mining loads, improve system resilience, and pave the way for cost-effective integration of cryptocurrency mining within renewable-rich smart grids.

3. System Modeling and Methodology

3.1. Power System Modeling

The power system under consideration integrates distributed renewable generation sources, specifically PV panels and wind turbines, combined with a diesel generator as a backup to ensure a continuous power supply. This hybrid system is designed to support both traditional urban loads and cryptocurrency mining operations, such as Bitcoin mining, which impose substantial and consistent energy demands.

The architecture of the system is configured to balance energy production from renewable sources with the operational needs of the mining farms. PV panels are used to harness solar energy, while wind turbines capture wind energy, both of which contribute to the system's base load. However, due to the intermittent nature of renewable energy sources—where solar energy is dependent on sunlight intensity and wind energy on wind speed—a diesel generator is introduced as a contingency measure to address the shortfall in power supply during low renewable generation periods.

3.1.1. PV System

The PV system is modeled based on solar radiation data and the geographic location of the installation site. The system's capacity is determined by the number and size of PV panels, as well as the efficiency of solar conversion. The energy output from the PV panels is calculated using real-time solar irradiance data, ensuring that the energy generation profile is accurately reflected in the model. The panels are connected to an inverter to convert direct current (DC) into alternating current (AC) to meet the load requirements of both urban consumers and cryptocurrency mining devices. Table 2 presents the technical and economic specifications of the solar panel used in this article for simulation purposes.

Table 2. Technical and Economic Specifications of PV Panel [60].

Parameter	Value
Name/Model	SunPower SPR-X22-370
Rated Output Power (kW)	0.37 kW (370 W)
Initial Investment Cost (USD/kW)	700–1000 USD/kW
Operation & Maintenance Cost (USD/kW/year)	10–20 USD/kW/year
Replacement Cost (USD/kW)	600–800 USD/kW
Nominal Operating Temperature (°C)	45–50 °C
Module Efficiency (%)	22.80%
Lifespan (years)	25 years
Tracking System	Fixed or Single-axis Tracking
Annual Degradation Factor (%)	0.25% per year
Temperature effects on power (%/°C)	−0.29%/°C
Nominal operating cell temperature (NOCT) (°C)	45 °C
Efficiency at standard test conditions (STC) (%)	22.80%
Derating Factor based on installation situation (%)	10–15% (Dust, Shading, etc.) *

* Note: In our study, we assumed that after each dust storm, the PV panels would be cleaned and washed if their efficiency was noticeably reduced. This practical assumption reflects common maintenance practices in dusty regions and ensures that the impact of dust deposition does not accumulate over time in our performance estimations.

3.1.2. Wind Turbine System

Wind turbines are included to complement solar generation, particularly during nighttime or cloudy periods when solar generation is minimal. The power output of the wind turbines is modeled based on wind speed data and turbine specifications, including rotor diameter, tower height, and conversion efficiency. The wind speed at the turbine's height is corrected using a wind shear model to account for varying conditions at different elevations. Like the PV system, the generated power is converted from DC to AC to be compatible with the grid and the mining farms.

The rated capacity of the wind turbine, which is an input parameter, does not affect the optimization process. However, varying the capacity of other equipment can impact the accuracy of the solution. Wind turbines are typically considered to have a rated capacity of 10 kW, while PV panels can be further modularized for greater flexibility.

Table 3 presents the technical and economic specifications of the wind turbine used in this article for simulation purposes.

Table 3. Technical and Economic Specifications of wind turbine [61].

Parameter	Value
Name/Model	Tumo-Int 10 kW Wind Turbine
Rated Output Power	10 kW
Capital Investment Cost	\$15,000–20,000 USD
Operation & Maintenance Cost	\$200–250 USD/year
Replacement Cost	\$15,000–18,000 USD
Lifespan	15–20 years
Cut-in Speed	3.0 m/s
Hub Height	10 m
Output Voltage	400 V (three-phase)
Availability Losses ¹	0%
Wake Effects Losses ²	5%
Turbine Performance Losses ³	10%
Electrical Losses ⁴	2%
Environmental Losses ⁵	2%
Curtailment Losses ⁶	2%
Other Losses ⁷	1%
Overall, Loss Factor	22%

Notes: 1. Availability Losses refer to downtime due to maintenance or other operational interruptions. A value of 0% indicates high reliability. 2. Wake Effects Losses are due to the turbulence created by other turbines in close proximity. 3. Turbine Performance Losses reflect inefficiencies in energy conversion at various wind speeds. 4. Electrical Losses typically occur in the wiring and connection systems. 5. Environmental Losses can include effects like bird and bat interactions or noise restrictions. 6. Curtailment Losses refer to reduced output due to grid demands or operational constraints. 7. Other Losses may include minor inefficiencies not covered by the different categories.

3.1.3. Diesel Generator Backup

Given the stochastic nature of renewable resources, the diesel generator serves as a backup source, ensuring system reliability during periods of low renewable generation. The generator's operation is triggered when the renewable output falls below a predefined threshold, and its fuel consumption is modeled based on load demand. The generator is primarily designed to operate intermittently, providing power during critical periods when renewable sources are insufficient. The inclusion of the diesel generator ensures that the cryptocurrency mining farms, which require consistent and high-power input, remain operational without interruption. Table 4 outlines the technical and financial details of the diesel generator utilized in this study for simulation purposes. Table 5 presents the emissions produced per liter of diesel fuel, providing key data for environmental impact assessment.

Table 4. Technical and Economic Specifications of Diesel Generator Unit [62].

Parameter	Value
Name/Model	Caterpillar C4.4 Diesel Generator
Rated Output Power (kW)	50 kW
Capital Investment Cost (USD)	Approx. 10,000–12,000
Replacement Cost (USD)	8000–10,000
Operation & Maintenance Cost (USD/kW/year)	15–25 USD, kW/year
Lifespan (h)	20,000–30,000 h
Government Diesel Fuel Price (USD/liter)	0.0047–0.0095
Global Diesel Fuel Price (USD/liter)	0.95
The following parameters are required in HOMER Pro	
Reference Generator Capacity	50 kW
Intercept Coefficient (L/h/kW rated)	0.23
Slope (L/h/kW output)	0.15
Carbon Monoxide (g/L of fuel)	0.02–0.04
Unburned Hydrocarbons (g/L of fuel)	0.01–0.03
Particulate Matter (g/L of fuel)	0.098
Proportion of Fuel Sulfur converted to PM (%)	0.15%
Nitrogen Oxides (g/L of fuel)	0.04–0.06

Table 5. Emissions per Liter of Diesel Fuel [63,64].

Pollutant	Amount (g/L of Fuel)
Carbon Dioxide (CO ₂)	2645.7
Carbon Monoxide (CO)	0.04
Unburned Hydrocarbons (HC)	0.03
Particulate Matter (PM)	0.098
Fuel Sulfur converted to PM	15%
Nitrogen Oxides (NO _x)	0.06

3.1.4. Storage System

The storage system in the power model serves to balance the intermittent energy output from renewable sources like solar and wind. It plays a crucial role in ensuring a consistent energy supply during periods when generation is low, such as cloudy or windless hours. Standard storage systems include battery energy storage systems (BESS), which can store surplus electricity generated during peak production periods and discharge it when needed. The effectiveness of the storage system is measured in terms of its capacity (in kWh) and ability to maintain grid stability. The storage system contributes to reducing the dependency on non-renewable backup systems, optimizing the use of renewable energy sources, and improving the system's overall energy efficiency. Table 6 summarizes the technical and financial characteristics of the Battery Storage Unit used in this research for simulation purposes. The data provided is based on the HOMER Pro hardware catalog.

Table 6. Technical and Economic Specifications of Battery Storage Unit [65–67].

Parameter	Value
Brand	kinetic battery model
Type	Lead Acid
Capital Investment Cost (USD)	300
Replacement Cost (USD)	300
Operation & Maintenance Cost (USD/year)	10
Lifespan (years)	10
Nominal Voltage (V)	12
Nominal Capacity (kWh)	1
Maximum Capacity (Ah)	83.4
Capacity Ratio	0.403
Rate Constant (1/h)	0.827
Roundtrip Efficiency (%)	0.8
Maximum Charge Current (A)	16.7
Maximum Discharge Current (A)	24.3
Maximum Charge Rate (A/Ah)	1
Throughput (kWh)	0.8
Site-Specific Input	
String Size	1
Initial State of Charge	100%
Minimum State of Charge	40%

3.1.5. Power Converter System

Power converters are vital in interfacing renewable energy systems with the grid or other load systems. The converter system typically converts the DC power generated by renewable sources (such as PV panels) into AC power, which is the standard form used in most power grids. Additionally, converters are used to manage bidirectional power flows, especially in systems incorporating storage devices, where the power needs to be

converted back and forth between DC and AC. A well-designed power converter system ensures minimal power losses during the conversion process, improving the efficiency of the overall power system. The converters also play a critical role in maintaining power quality by regulating voltage and frequency to ensure grid compatibility. Table 7 presents the technical and economic specifications of the diesel generator used in this article for simulation purposes.

Table 7. Technical and Economic Specifications of the Power Converter (Inverter/Rectifier) Unit [68].

Parameter	Value
Name/Model	SMA Sunny Tripower 5.0 Smart Energy—Hybrid
Capacity	56 kW (if $\cos \varphi = 1$)
Nominal power	56,000 VA
Lifespan (years)	10
Converter and Rectifier Efficiency	98%
Capital Investment Cost (USD/kW)	\$42.5
Replacement Cost (USD/kW)	\$42.5
Operation & Maintenance Cost (USD/kW/year)	\$15
Storage system	DC coupling
Number of MPPTs	2 pieces
MPP voltage min	250.0 volts
MPP voltage max	800.0 volts
Dimensions in mm	L: 173 mm W: 500 mm H: 598 mm
Weight	31 kg

We acknowledge that lower-cost or different models are available locally (e.g., in the Iranian market, where Chinese equivalents like Yingli solar panels or local wind turbines are more accessible.); however, for the simulations in this study we intentionally used widely known and well-documented components (SunPower SPR-X22-370 PV from SunPower Corporation, San Jose, CA, USA; Tumo-Int 10 kW wind turbine from Tumo Int Corporation Limited, Guangzhou, Guangdong, China; Caterpillar C4.4 diesel generator from Caterpillar Inc., Peoria, IL, USA; the battery unit from the HOMER Pro catalog, which includes generic models from various manufacturers curated by UL Solutions, Boulder, CO, USA; and the SMA Sunny Tripower inverter from SMA Solar Technology AG, Niestetal, Hesse, Germany) for several practical reasons. First, global datasheets and verified price ranges for these models are publicly available, which makes our assumptions reproducible and easy for reviewers and future researchers to check. Second, these brands and models offer higher build quality and proven reliability, so their performance data (efficiency, degradation, failure rates) are more stable and lead to more realistic simulation results. Third, using established international products creates a solid baseline for future studies: other researchers can repeat or extend our work using the same reference equipment. Finally, from an economic point of view, we note that even when using relatively expensive, high-quality equipment, the internal rate of return (IRR) values obtained under subsidized energy scenarios remained reasonable. If cheaper, lesser-known devices were used instead (such as those commonly sourced in Iran from Chinese manufacturers), the initial capital cost might fall but lifetime efficiencies, replacement needs, and failure risks would likely increase—producing economic indicators (e.g., IRR) that look better on paper but are not robust or broadly reproducible unless those exact local parts are accessible to other researchers. For these reasons, the chosen models represent a conservative and verifiable selection that balances technical realism and reproducibility.

3.1.6. Cryptocurrency Mining Load

Cryptocurrency mining farms impose a substantial load on the system, as the mining process requires continuous high-power consumption for running mining rigs. The energy consumption of each mining device is included in the load profile, with adjustments for efficiency losses and cooling requirements. The load demand is modeled to reflect both daily operational fluctuations and long-term trends in cryptocurrency mining intensity. Since the mining devices operate continuously, the system's capacity must be sufficient to meet this load regardless of the availability of renewable resources. Table 8 presents the technical specifications of cryptocurrency mining devices with the best performance and profitability in the market as of September 2024 [69].

3.1.7. Temporal Variability in Urban and Rural Electricity Loads

Accurate modeling of electricity demand in urban areas requires accounting for the temporal dynamics that characterize residential and commercial consumption. In contrast to static load assumptions, the urban load in this study was modeled using real-world data, encompassing both hourly and monthly variations over a typical year.

The Hourly Load profile, derived from empirical data, exhibits clear daily peaks corresponding to morning (approximately 7:00–9:00) and evening (around 18:00–22:00) usage, which align with typical household and business activity patterns. Minimum demand is observed during late-night hours (midnight to 5:00), indicating reduced occupancy and operational activity in residential zones.

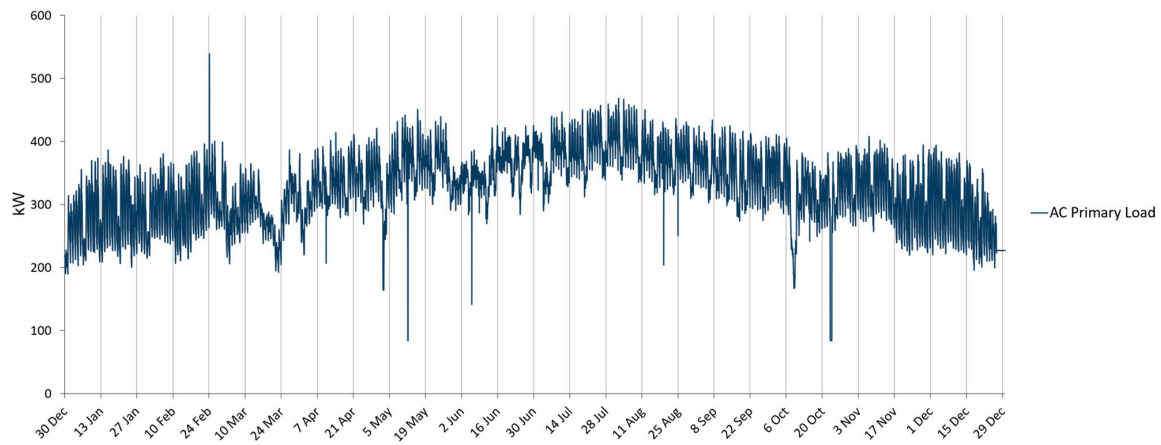
Seasonal variations are also significant, as illustrated in the Monthly Load graph. Peak demand occurs during the summer months—particularly July and August—mainly due to the extensive use of air conditioning systems. A secondary peak is visible in January, reflecting heating loads in colder periods. Conversely, the spring and autumn months (e.g., April and October) exhibit relatively lower energy consumption due to milder weather conditions.

The Load Profile and CDF (Cumulative Distribution Function) Load diagrams further support the statistical distribution of urban electricity demand, indicating the prevalence of mid-range consumption levels with occasional high-load instances likely associated with simultaneous HVAC, lighting, and appliance use.

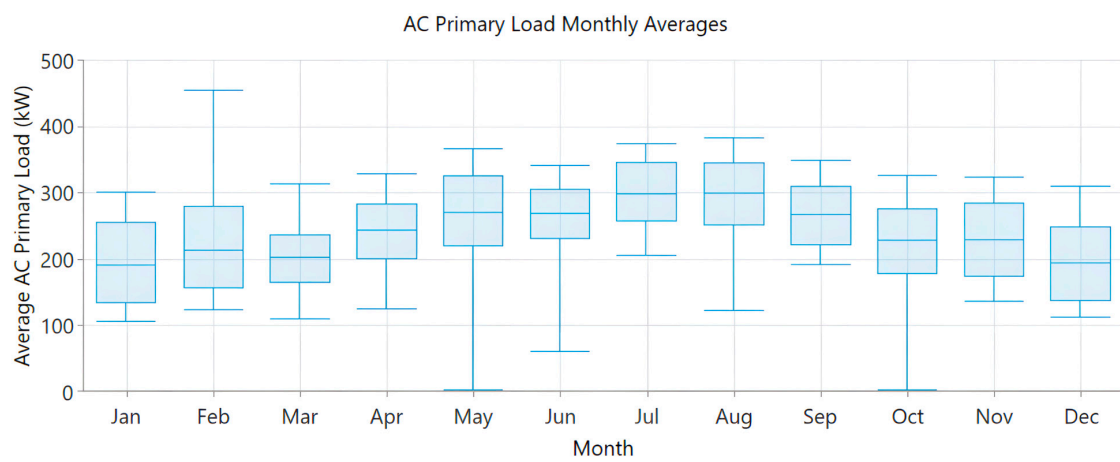
By integrating this granular time-series data into the system model, the simulation accurately reflects real-life load dynamics, enhancing the robustness of energy optimization strategies and enabling a more precise evaluation of performance metrics under variable demand conditions. All discussed patterns and load variations are illustrated in detail in the compiled visual dataset provided in Figure 2.

Table 8. Technical and Economic Specifications of the Best Cryptocurrency Mining Devices [69].

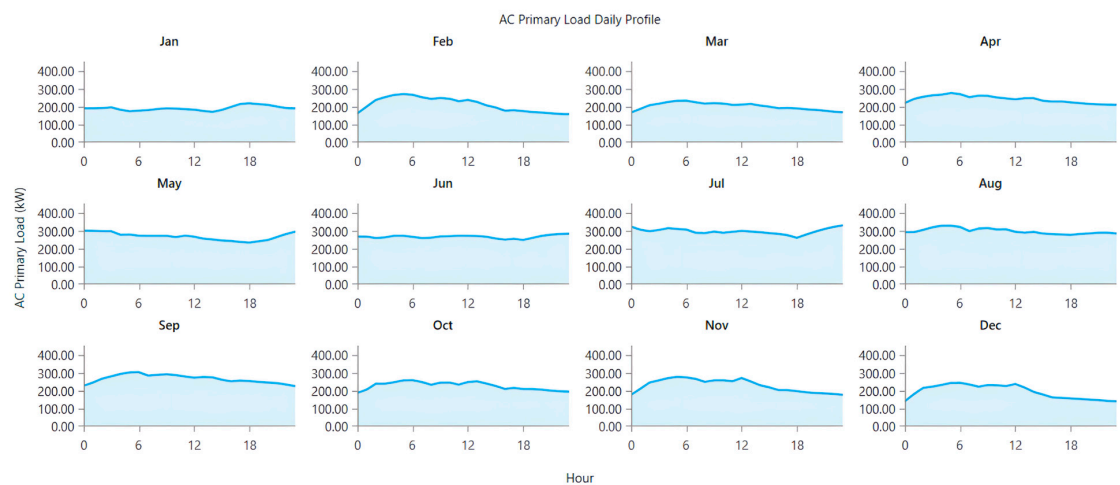
Row	Manufacturer	Miner Model	Release	Hashrate	Power	Efficiency	Top Minable Coins Based on Profit	Algorithm	Income/Day	Electricity/Day	Profit/Day
1	Bitmain	Bitmain Antminer AL1 Pro (16.6 Th)	September 2024	16.6 Th/s	3730 W	0.225 j/Gh	ALPH (Alephium)	Blake3	\$50.67	\$7.16	\$43.51
2	Bitmain	Bitmain Antminer AL1 (15.6 Th)	July 2024	15.6 Th/s	3510 W	0.225 j/Gh	ALPH (Alephium)	Blake3	\$47.62	\$6.74	\$40.88
3	IceRiver	IceRiver AL3	October 2024	15 Th/s	3500 W	0.233 j/Gh	ALPH (Alephium)	Blake3	\$45.79	\$6.72	\$39.07
4	Bitmain	Bitmain Antminer S21e XP Hyd 3U	December 2024	860 Th/s	11,180 W	13 j/Th	BTC (Bitcoin)	SHA-256	\$40.43	\$21.47	\$18.96
5	Bitmain	Bitmain Antminer L9 (16 Gh)	May 2024	16 Gh/s	3360 W	0.21 j/Mh	DOGE (Dogecoin) & LTC (Litecoin)	Scrypt	\$25.59	\$6.45	\$18.31
6	Bitmain	Bitmain Antminer L9 (17 Gh)	May 2024	17 Gh/s	3260 W	0.21 j/Mh	DOGE (Dogecoin) & LTC (Litecoin)	Scrypt	\$24.92	\$6.85	\$19.45
7	ElphaPex	ElphaPex DG 1+	April 2024	14 Gh/s	3920 W	0.28 j/Mh	DOGE (Dogecoin) & LTC (Litecoin)	Scrypt	\$22.39	\$7.53	\$14.14
8	Bitmain	Bitmain Antminer S21 XP Hyd (473 Th)	November 2024	473 Th/s	5676 W	12 j/Th	BTC (Bitcoin)	SHA-256	\$22.24	\$10.90	\$11.34
9	Bitmain	Bitmain Antminer KS5 Pro (21 Th)	March 2024	21 Th/s	3150 W	0.15 j/Gh	KAS (Kaspa)	KHeavyHash	\$17.66	\$6.05	\$13.39
10	Bitmain	Bitmain Antminer KS5 (20 Th)	March 2024	20 Th/s	3000 W	0.15 j/Gh	KAS (Kaspa)	KHeavyHash	\$16.30	\$5.76	\$12.75
11	ElphaPex	ElphaPex DG 1	March 2024	11 Gh/s	3420 W	0.311 j/Mh	DOGE (Dogecoin) & LTC (Litecoin)	Scrypt	\$16.08	\$6.57	\$10.46



(a)

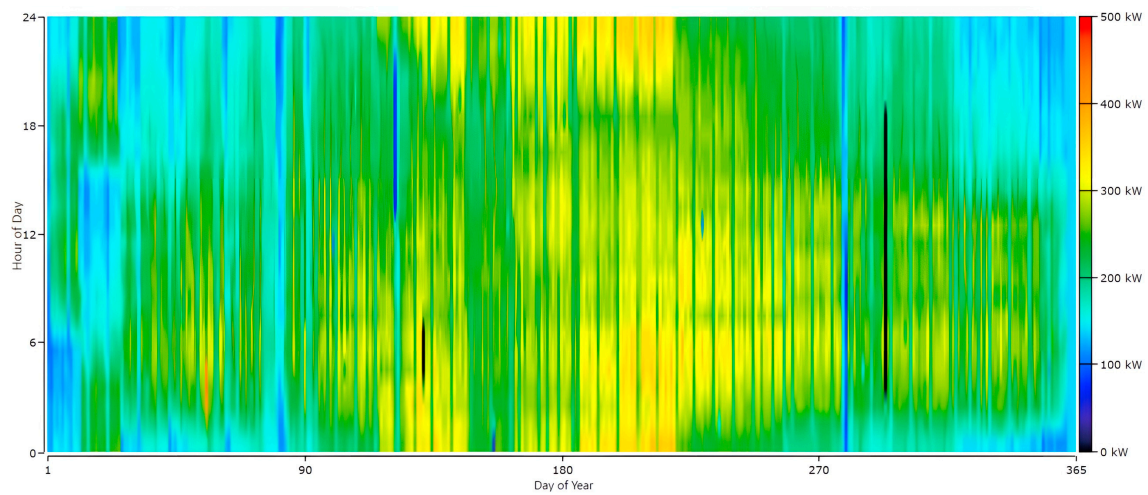


(b)

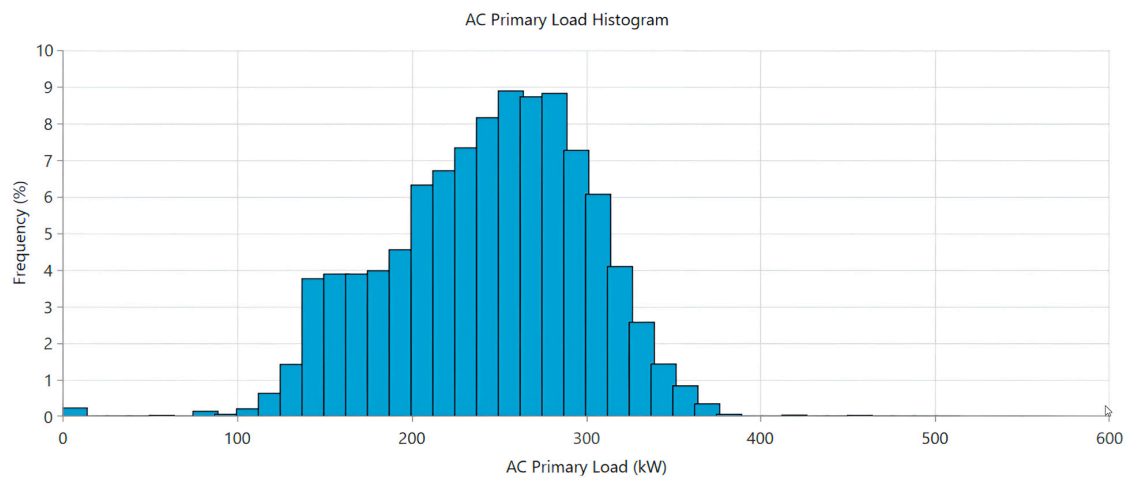


(c)

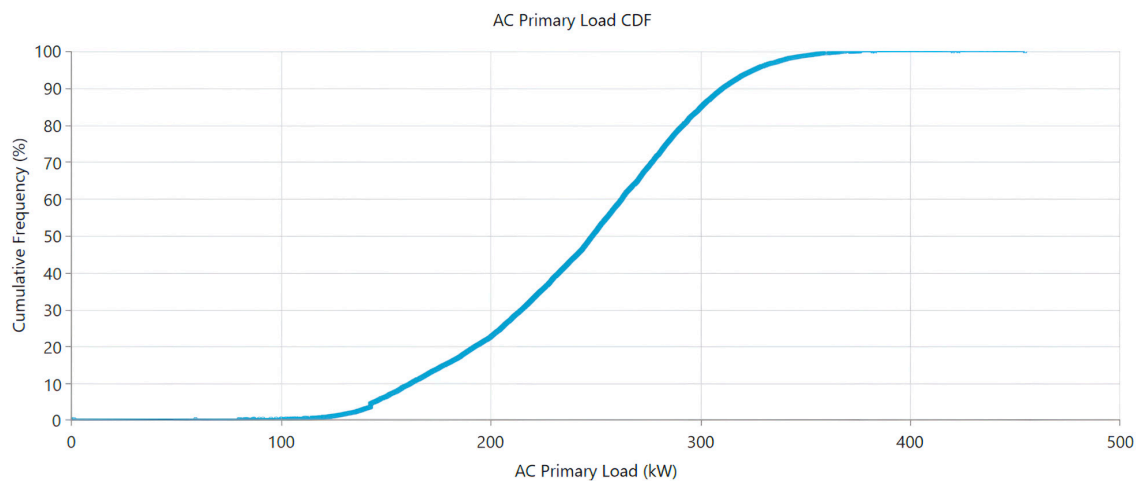
Figure 2. Cont.



(d)



(e)



(f)

Figure 2. Temporal Load Characteristics in Urban Areas Used in the Simulation Model; (a) AC primary load (kW)—hourly profile; (b) AC primary load (kW)—monthly average; (c) AC primary load (kW)—user-defined daily profile; (d) AC primary load (kW)—density map (DMap); (e) AC primary load (kW)—histogram distribution; (f) AC primary load (kW)—cumulative distribution function (CDF).

3.2. Optimization Model

The optimization of the hybrid energy system, which integrates distributed renewable generation sources (PV panels, wind turbines) with cryptocurrency mining farms and a backup diesel generator, is critical for ensuring both technical efficiency and economic viability. The primary objective of the optimization model is to minimize the total cost of energy (COE) while satisfying the system's power demand, maintaining operational reliability, and maximizing the use of renewable energy sources. The optimization problem is formulated to address various system constraints, including the stochastic nature of renewable resources, energy storage limitations, and the continuous power demands of cryptocurrency mining farms.

3.2.1. Problem Formulation

The optimization problem can be mathematically formulated as a mixed-integer non-linear programming (MINLP) model that seeks to minimize the Net Present Cost (NPC) of the hybrid energy system. The total NPC includes the capital cost, operating and maintenance costs (O&M), replacement costs, and fuel costs associated with the diesel generator. The decision variables in the optimization model include the following:

The size and number of PV panels and wind turbines.

The capacity of the diesel generator and energy storage (if applicable).

The dispatch strategy for operating the diesel generator.

The operational profile of the cryptocurrency mining farms.

The objective function can be expressed as

$$NPC = C_{capital} + C_{O\&M} + C_{replacement} + C_{fuel} \quad (1)$$

where

$C_{capital}$ is the total capital cost of the renewable energy components, diesel generator, cryptocurrency miners, rental, human resources, and internet consumption.

$C_{O\&M}$ is the operational and maintenance cost, which includes servicing, repairs, and other recurring expenses.

$C_{replacement}$ represents the cost of replacing components like batteries or inverters over the system's lifecycle.

C_{fuel} is the fuel cost incurred by the operation of the diesel generator, which depends on fuel consumption and price.

The initial energy capital cost is the total cost of all components at the start of operation. In a system consisting of batteries, wind turbines, solar panels, a diesel generator, and a converter, the initial capital cost can be calculated using [70].

$$C_{capital} \text{ (for energy)} = N_b C_b + \frac{P_{st}}{P_{sb}} C_s + \frac{P_{wt}}{P_{wb}} C_w + P_{gt} C_{gref} + \frac{P_c}{P_{cb}} C_c \quad (2)$$

N_b is the number of batteries, C_b is the price of each battery in dollars, P_{st} is the total power generated by all solar panels in kilowatts, P_{sb} is the power generated by a single solar panel in kilowatts, C_s is the price of each solar panel in dollars, P_{wt} is the total power generated by all wind turbines in kilowatts, P_{wb} is the power generated by a single wind turbine in kilowatts, C_w is the price of the wind energy system per wind turbine in dollars, P_{gt} is the total power generated by the diesel generator in kilowatts, C_{gref} is the price of a 1-kilowatt generator in dollars, P_c is the converter capacity in kilowatts, P_{cb} is the base converter capacity in kilowatts, and C_c is the price of the base converter in dollars.

The initial capital cost for mining devices can be calculated using (3).

$$C_{capital} (\text{Cost Of Miner}) = N_{Miners} \times C_{Miner} \quad (3)$$

In this equation, N_{Miners} represents the number of mining devices, and C_{Miners} is the cost of each device, which together constitute the initial capital cost.

The operating costs refer to the total costs associated with the operation of all system components in a year. The total operating cost can be calculated using (4) [70].

$$TOC = \sum_{i=1}^n C_{OM,i} + C_f + \sum_{i=1}^n C_{R,i} - \sum_{i=1}^n C_{S,i} \quad (4)$$

where n is the number of system components, $C_{OM,i}$ is the annual operation and maintenance cost for each component of the system, C_f is the total annual fuel cost, $C_{R,i}$ is the annual replacement cost for each system component, and $C_{S,i}$ is the salvage value of the i component.

In HOMER Pro software, components depreciate linearly. The salvage value of a component is directly proportional to its remaining lifespan. The replacement cost is considered to account for the economic impact of using components with a lifespan shorter than the project's duration.

Also, the NPC of a hybrid energy system could be defined according to (5) [70].

$$NPC(USD) = \frac{TAC}{CRF} \quad (5)$$

where TAC is the total annual cost, CRF and is the capital recovery factor.

The capital recovery factor would be determined as (6) [70].

$$CRF(USD) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (6)$$

where N is the defined project lifetime (in years), and i is the annual interest rate.

Equation (7) calculates the NPC of the mining operation for each year based on year zero.

$$NPC_{Cryptocurrencies} = \frac{A}{(1+i)^n} \quad (7)$$

where n is the number of project years, i is the real discount rate, and A is the total annual mining revenue.

The payback period is an essential parameter for project selection. It is the time at which the total cost of the project is recovered through the project's cash flows, indicating that the project will be profitable after this period. The IRR is equivalent to the rate of return an investor can achieve by investing in a business plan. This metric is the most widely used indicator for the financial feasibility and comparison of multiple business plans. The calculation of the IRR is closely related to the net present value (NPV), where the NPV of the project is set to zero to determine the discount rate that yields the IRR. Equation (8) shows the internal rate of return indicator [71].

$$\sum_{n=0}^N \frac{C_n}{(1+IRR)^n} = 0 \quad (8)$$

where N is the project lifetime in years, and C_n is the cash flow for year n .

3.2.2. Constraints

The optimization model is subject to several key constraints to ensure the feasibility of the solution:

- **Power Balance Constraint:** The total power generated by the PV panels, wind turbines, and diesel generator must meet the total load demand (urban load + cryptocurrency mining load, based on scenarios) at all times:

$$P_{PV} + P_{wind} + P_{diesel} \geq P_{load} \quad (9)$$

- **Renewable Energy Generation Constraint:** The power generated by renewable energy sources is limited by their respective maximum generation capacities, which depend on real-time solar irradiance and wind speed:

$$0 \leq P_{PV} \leq P_{PV}^{max}, 0 \leq P_{wind} \leq P_{wind}^{max} \quad (10)$$

- **Diesel Generator Operation Constraint:** The diesel generator is only utilized when the renewable energy generation is insufficient. Its operation is constrained by fuel availability and the generator's capacity:

$$0 \leq P_{diesel} \leq P_{diesel}^{max} \quad (11)$$

- **Battery Storage Constraints (if applicable):** If energy storage systems are incorporated, additional constraints govern the charging and discharging of the batteries, ensuring that they operate within their rated capacities:

$$0 \leq E_{storage} \leq E_{storage}^{max} \quad (12)$$

- **Cryptocurrency Mining Load Constraint:** Cryptocurrency mining farms demand continuous and stable power. This load must always be met without interruptions and is calculated as (13) [72]:

$$P_{mining} = \frac{NH \times EoH}{10^{12}} \quad (13)$$

where P is the power (in megawatts), NH is the network hash rate (in mega hashes per second), and EoH is the hardware efficiency (energy consumed by hardware per terahash).

- **Cooling Power for Miners:** Cryptocurrency mining rigs generate significant heat due to their high computational demands. This heat must be effectively managed to prevent hardware damage and maintain optimal performance. Therefore, the cooling system must provide sufficient power to cool the mining devices continuously. The required cooling power depends on the total heat load generated by the miners, which is a function of the number of devices, their power consumption, and their efficiency. The optimization model ensures that the energy allocated for cooling is sufficient to maintain safe operating temperatures without compromising the energy available for other loads, such as cryptocurrency mining or urban energy demands.
- To calculate the cooling power required for cryptocurrency miners, you can use the following formula based on the heat generated by the mining devices [73–75]:

$$P_{cooling} = \frac{P_{miners} \times (1 - \eta_{efficiency})}{COP} \quad (14)$$

where $P_{cooling}$ is the cooling power required (in kilowatts, kW), P_{miners} is the total power consumption of the mining equipment (in kW), and $\eta_{efficiency}$ is the efficiency of the mining devices (fractional value between 0 and 1; for instance, 0.9 for 90% efficient devices), COP is the Coefficient of Performance of the cooling system (typically 3 to 5

for efficient cooling systems).

Explanation:

1. The power consumed by the miners (P_{miners}) is mostly converted to heat. The efficiency of the mining devices reduces the amount of power that needs to be removed as heat.
2. The Coefficient of Performance (COP) represents how effectively the cooling system converts power input into cooling power. A higher COP means more efficient cooling.
3. This formula allows you to estimate the cooling power required based on the power consumption of the miners, their efficiency, and the cooling system's performance.

- **Optimization Techniques:**

The optimization process is implemented using the HOMER Pro software, which is specifically designed for hybrid energy systems. HOMER Pro performs simulations to evaluate the performance of different system configurations by varying the sizes and combinations of PV panels, wind turbines, and diesel generators. HOMER Pro's optimization algorithm identifies the configuration that minimizes the NPC while meeting technical and economic constraints.

The software performs detailed hourly simulations, taking into account variations in renewable energy availability, load profiles, and fuel prices. It optimizes the balance between renewable energy generation and diesel backup, ensuring that the system operates at the lowest possible cost while meeting load requirements consistently.

3.2.3. Sensitivity Analysis

HOMER Pro also allows for sensitivity analysis to test how variations in critical parameters, such as fuel prices, solar and wind availability, and equipment costs, impact the system's overall performance. This helps ensure the robustness of the optimized configuration and assists decision-makers in planning for future uncertainties. Sensitivity analysis provides insights into the trade-offs between costs and system reliability under different scenarios.

3.3. Simulation Framework

The simulation framework for the hybrid energy system, which includes distributed renewable generation sources (PV panels and wind turbines), cryptocurrency mining farms, and a backup diesel generator, is designed to analyze and optimize the system's performance under varying operational conditions. The HOMER Pro (Hybrid Optimization of Multiple Energy Resources) software is utilized as the primary tool for simulating and optimizing the system. The framework consists of key components, including the tools used, data inputs, and critical assumptions that guide the simulation process.

3.3.1. Tools

The primary tool employed for the simulation is HOMER Pro, which is specifically designed for the modeling and optimization of microgrids and hybrid energy systems. HOMER Pro simulates various combinations of energy sources and evaluates the feasibility of each configuration based on both technical performance and economic cost. It allows for hourly simulations over a specified time horizon, typically one year, to account for variations in renewable energy generation and load demand.

HOMER Pro's features include the following:

- **Energy production modeling** for renewable sources such as solar panels and wind turbines.

- **Dispatch strategy optimization** to control the operation of the diesel generator and manage energy storage (if applicable).
- **Economic analysis** that evaluates the total COE, NPC, and ROI.
- **Sensitivity analysis** to assess how variations in key parameters affect the system's performance and cost.

3.3.2. Data Inputs

To accurately simulate the system's performance, a range of input data is required. These data inputs are essential for reflecting the real-world conditions under which the hybrid system operates:

- **Solar and Wind Data:** Real-time solar irradiance and wind speed data for the geographic location where the system is deployed are critical inputs. These datasets are typically gathered from local meteorological stations or publicly available databases such as NASA's Surface Meteorology and Solar Energy dataset. The data is used to estimate the energy output of the PV panels and wind turbines throughout the year.
- **Load Profiles:** Two distinct load profiles are considered in the simulation:
 1. **Urban Load:** This reflects the typical electricity consumption of urban residential and commercial users in the system's service area. The load profile varies over the course of the day, with peak loads typically occurring during the day and early evening.
 2. **Cryptocurrency Mining and Cooling Loads:** Cryptocurrency mining farms operate continuously and demand a stable, high-power input. The load is modeled as a constant demand throughout the simulation period to reflect the energy-intensive nature of cryptocurrency mining operations.
- **System Components Data:** Specifications for all system components, including PV panels, wind turbines, and the diesel generator, are input into HOMER Pro. This includes technical parameters such as rated capacity, efficiency, lifetime, and operational costs. Fuel consumption rates and fuel prices are also key inputs for the diesel generator.
- **Economic Data:** Capital costs, O&M costs, and replacement costs for all components are input into the model. Additionally, the discount rate and inflation rate are specified to calculate the system's NPC over its operational lifetime. HOMER Pro takes into account both local and U.S. diesel fuel prices to accurately represent real-time operational costs.

3.3.3. Assumptions

Several assumptions are made to simplify the simulation process and ensure realistic outputs:

- **Constant Cryptocurrency Mining Load:** It is assumed that the cryptocurrency mining load remains constant throughout the simulation period. This assumption reflects the continuous operation of mining devices, which require a stable and uninterrupted power supply.
- **Solar and Wind Intermittency:** Solar and wind energy generation is subject to natural fluctuations based on weather patterns. It is assumed that the solar and wind data provided represent typical annual conditions and that extreme variations (e.g., prolonged cloudy or windless periods) are rare.
- **Backup Diesel Generator Operation:** The diesel generator is assumed to operate only during periods when renewable generation (solar and wind) is insufficient to meet the system's load. It is dispatched based on an optimized strategy that minimizes fuel consumption and operational costs.

- **Component Lifetimes and Replacement Cycles:** The lifetimes of the PV panels, wind turbines, and diesel generators are assumed based on manufacturer specifications. The model incorporates replacement cycles for components that are expected to fail or degrade over time, such as batteries or inverters (if used).
- **No Grid Connection:** The system is assumed to operate as an off-grid, stand-alone hybrid system. No electricity is imported from or exported to a central utility grid, making the system fully independent in its energy generation and consumption. It should be noted that the “No Grid Connection” assumption applies to the baseline system design and all standard scenarios in this study. However, in Scenario 3 (Section 5.2.3), a special case is introduced where limited electricity sales to the national grid are modeled. This exception is included only for comparative and economic evaluation during peak summer months and does not change the fundamental off-grid nature of the system in other scenarios.

3.3.4. Simulation Process

The simulation is run on an hourly basis over a period of five years. HOMER Pro evaluates the energy balance at each hour, ensuring that the total power generated by the renewable sources and the diesel generator meets the combined load demand. The software continuously adjusts the dispatch strategy of the diesel generator to minimize fuel usage and optimize the overall COE.

HOMER Pro also performs a comprehensive economic analysis by calculating the total NPC, levelized cost of energy (LCOE), and payback period for the hybrid system. Sensitivity analyses are conducted by varying key inputs, such as fuel prices, renewable energy availability, and equipment costs, to understand how these factors impact the system’s economic performance.

4. Proposed Optimization Algorithm

4.1. Algorithm Design

The proposed optimization algorithm is designed to determine the optimal configuration and operational strategy for the hybrid energy system, which includes distributed renewable energy sources (PV panels and wind turbines), cryptocurrency mining farms, and a backup diesel generator. The goal of the algorithm is to minimize the total system cost, including capital expenditures, operational and maintenance costs, and fuel consumption, while ensuring a reliable power supply for both urban loads and cryptocurrency mining operations.

The optimization problem is complex due to the stochastic nature of renewable energy sources and the continuous energy demands of cryptocurrency mining. Therefore, the algorithm is designed to adjust the energy supply from renewable sources dynamically and the backup diesel generator based on real-time energy demand and resource availability.

4.1.1. Algorithm Structure

The algorithm follows a hybrid optimization approach, combining simulation-based analysis (via HOMER Pro software) with rule-based dispatch strategies. The key steps in the algorithm are as follows:

- **Step 1: Input Data Initialization**

The algorithm begins by collecting and initializing all necessary input data, including the following:

1. Renewable energy data (solar irradiance and wind speed).
2. System load profiles (urban and cryptocurrency mining load).

3. Technical specifications and costs for PV panels, wind turbines, diesel generator, storage system, and power converter system.
4. Minimum operating reserve (percent/solar output power)
5. Energy system project lifetime (years)
6. Economic data such as capital cost, operational cost, fuel prices, nominal discount rate (percent) and inflation rate (percent).

This data is used to model the system components and energy production capacities under various operating conditions.

Table 9 summarizes the technical and economic variables used in this research for simulation purposes.

Table 9. Significant technical and economic variables.

Parameter	Value Global (in USA)
Energy system project lifetime (years)	5
Nominal Discount Rate in 2024 (percent)	~5.4% [76]
Inflation rate in 2024 (percent)	~4.6% [76]
Minimum operating reserve (percent/solar output power)	80%
Minimum operating reserve (percent/wind output power)	50%
Minimum load Ratio (percent/Diesel Generator output power)	25%

• Step 2: Renewable Energy Generation Calculation

For each hour of the simulation period (typically one year), the algorithm calculates the energy output from the PV panels and wind turbines. The PV energy generation is computed using the following Equations (15) and (16) [70,77,78]:

$$P_{PV} = A_{PV} \times \eta_{PV} \times G_t \quad (15)$$

where P_{PV} is the power generated by the PV panels, A_{PV} is the area of the PV array, η_{PV} is the efficiency of the panels, and G_t (in W/m^2) is the solar irradiance.

$$P_{PV} = f_{PV} Y_{PV} \frac{I_t}{I_s} \quad (16)$$

where f_{PV} is the PV derating factor, Y_{PV} is the rated capacity of the PV system in kilowatts, I_t is the global solar irradiance on the surface of the PV panel in kilowatts per square meter, and I_s is the standard irradiance used for rating the PV array capacity, which is 1 kilowatt per square meter.

The PV panel derating factor generally results from factors such as dust on the panel surface, shading, panel degradation, and temperature rise.

The output power of a wind turbine is calculated as (17) [70].

$$P_{wind} = \frac{1}{2} \times \rho \times A \times v^3 \times \eta_{wind} \quad (17)$$

where P_{wind} is the power generated by the wind turbines, ρ is the air density, A is the swept area of the turbine, v is the wind speed, and η_{wind} is the efficiency of the turbine.

Battery dynamics are defined by the state of charge (SOC) [79]:

$$SOC(t) = SOC_0 + \frac{1}{Q_{rated}} \int_0^t \eta \times I(\tau) d\tau \quad (18)$$

Here, $SOC(t)$ represents the battery's state of charge at time t , typically expressed as a percentage or a ratio (0 to 1) of the remaining capacity relative to the nominal capacity. SOC_0 is the initial state of charge at time rated $t = 0$. Q_{rated} denotes the rated capacity of

the battery, measured in ampere-hours (Ah), which indicates the total charge the battery can store when fully charged. $I(\tau)$ is the current flowing into (positive for charging) or out of (negative for discharging) the battery at time τ , measured in amperes (A). η represents the coulombic efficiency, a dimensionless factor (between 0 and 1) that accounts for energy losses during charging and discharging processes, with $\eta = 1$ indicating no losses. The integral $\int_0^t \eta \times I(\tau) d\tau$ calculates the net charge accumulated or depleted over time, adjusted for efficiency, and is divided by Q_{rated} rated to normalize the SOC relative to the battery's capacity.

In addition to the state of charge (SOC) dynamics, a rule-based control logic was developed to manage the battery's charging and discharging behavior, with the primary objectives of maximizing renewable energy utilization and minimizing diesel generator dependency. The control strategy is based on the net power balance between the load and renewable generation [80]:

$$P_{net} = P_{load} - P_{PV} \quad (19)$$

where P_{load} is the total system load (including mining and non-mining loads), and P_{PV} is the PV output. When $P_{net} < 0$, indicating an energy surplus, and the battery's SOC is below the upper limit ($SOC < SOC_{max} = 1.0$), the battery is charged according to the following [80,81]:

$$P_{charge} = \min(-P_{net}, P_{charge}^{max}) \quad (20)$$

Conversely, when $P_{net} > 0$, indicating a generation deficit, and $SOC > SOC_{min} = 0.4$, the battery is discharged as [80,81]:

$$P_{discharge} = \min(P_{net}, P_{discharge}^{max}) \quad (21)$$

The SOC is then updated at each time step according to the following equation [80,81]:

$$SOC_{t+1} = SOC_t + \frac{\eta_{ch} \cdot P_{charge} \cdot \Delta t - \frac{P_{discharge} \cdot \Delta t}{\eta_{dis}}}{C_{batt}} \quad (22)$$

where η_{ch} and η_{dis} are the charging and discharging efficiencies, Δt is the time step C_{batt} is the battery's rated capacity (kWh).

Charging logic:

$$P_{charge} = P_{renewable} - P_{load,total} \quad (23)$$

This condition ensures that the battery only charges when renewable generation exceeds the total load, including mining, and as long as $SOC < 100\%$

Discharging logic:

$$P_{discharge} = P_{load,total} - P_{renewable} \quad (24)$$

The battery discharges to meet demand when renewable generation is insufficient, provided that $SOC > 40\%$. These SOC limits (initial SOC = 100%, minimum SOC = 40%) were defined based on the technical characteristics listed in Table 6.

This logic aligns with HOMER Pro's default load-following dispatch strategy, which prioritizes battery charging from excess renewable energy and defers diesel generation until storage resources are depleted [82–84]. In scenarios where the battery cannot meet the net demand and diesel fuel is costly or limited, the diesel generator provides backup, ensuring energy balance and supply reliability.

- **Step 3: Non-Renewable Energy Generation Calculation**

For each hour of the simulation period (typically one year), the algorithm calculates the energy output from the diesel generator. The output power of a wind turbine is calculated as (25) [85–88].

$$E_{out} = P_{gen} \times t \times \eta \quad (25)$$

where E_{out} is the energy output of the diesel generator (in kilowatt-hours, kWh), P_{gen} is the rated power capacity of the generator (in kilowatts, kW), t is the time of operation (in hours, typically 1 h for hourly simulations), η is the efficiency of the diesel generator (a fraction between 0 and 1).

The diesel generator's fuel consumption is modeled using a fuel efficiency curve that correlates power output with fuel usage. This relationship is calculated based on the following [89–92]:

$$F_{diesel}(t) = a \times P_{diesel}(t) + b \quad (26)$$

where $F_{diesel}(t)$ is the fuel consumption at time t , $P_{diesel}(t)$ is the power output of the diesel generator, a and b are generator-specific constants.

- **Step 4: Load Demand Assessment**

The algorithm assesses the total system load for each hour, which includes the urban load and the cryptocurrency mining load. The cryptocurrency mining load is modeled as a constant demand due to the continuous operation of mining equipment. The total load is given by

$$P_{load}(t) = P_{urban}(t) + P_{mining} + P_{cooling} \quad (27)$$

where $P_{urban}(t)$ is the urban load at time t , P_{mining} is the constant cryptocurrency mining load, and $P_{cooling}$ represents the constant cooling power required to operate mining equipment.

- **Step 5: Energy Balance and Diesel Generator Dispatch**

The algorithm calculates the energy balance for each hour by comparing the total renewable energy generation with the load demand. If the renewable generation is insufficient to meet the load, the diesel generator is dispatched to cover the shortfall. The diesel generator operates based on a rule-based dispatch strategy, which aims to minimize fuel consumption while ensuring the load is met.

The dispatch decision is governed by the following rules:

1. If $P_{PV}(t) + P_{wind}(t) \geq P_{load}(t)$, then no diesel generation is required.
2. If $P_{PV}(t) + P_{wind}(t) < P_{load}(t)$, then the diesel generator is activated to supply the deficit $P_{diesel}(t)$.

- **Step 6: Economic Analysis**

For each configuration of the system (i.e., different numbers and capacities of PV panels, wind turbines, and diesel generators), the algorithm computes the total COE and the NPC. The NPC is the sum of capital, operational, and replacement costs, along with fuel costs. The COE is calculated by dividing the total annualized cost by the total energy generated [35]:

$$COE = \frac{C_{total}}{E_{total}} \quad (28)$$

where C_{total} is the total annualized cost, and E_{total} is the total energy generated by the system.

- **Step 7: Optimization and Selection of Best Configuration**

The algorithm iteratively simulates various system configurations and selects the one that minimizes the NPC while satisfying the power demand and ensuring system reliability. The optimal configuration is determined based on a combination of technical performance (energy balance, reliability) and economic performance (COE, ROI).

4.1.2. Flowchart of the Algorithm

Figure 3 shows a flowchart that outlines the sequential steps involved in the proposed optimization algorithm:

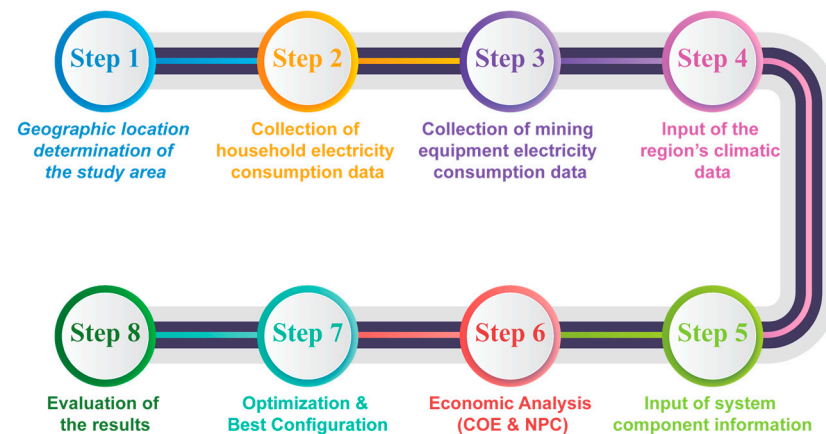


Figure 3. Step-by-step process of the proposed optimization algorithm.

4.2. Innovation and Advantages

The optimization algorithm proposed in this study introduces significant advancements over existing models, particularly in its application to energy systems integrated with cryptocurrency mining operations. Unlike traditional energy optimization approaches that primarily focus on cost or energy efficiency, this algorithm accounts for the fluctuating and high-energy demands of cryptocurrency mining while ensuring optimal resource allocation from both renewable and non-renewable energy sources. Below, we highlight the innovations and key benefits of the proposed approach.

4.2.1. Integration of Cryptocurrency Mining with Renewable Energy Systems

One of the most critical innovations is the inclusion of cryptocurrency mining as a substantial load in the energy system. Existing algorithms largely neglect or underrepresent the energy demands of mining operations. By contrast, our model directly incorporates the variable energy consumption patterns of cryptocurrency mining and synchronizes it with renewable energy outputs from PV systems and wind turbines. This integration results in a balanced energy portfolio, maximizing the use of renewable energy and minimizing reliance on fossil fuels.

4.2.2. Dynamic Energy Management Based on Real-Time Data

Our algorithm stands out for its capacity to dynamically manage energy resources based on real-time data inputs. It leverages weather forecasts for solar irradiance and wind speed to predict renewable energy availability and adjusts the energy supply to cryptocurrency mining operations accordingly. This dynamic adjustment optimizes energy use by minimizing waste and ensuring that the mining process operates primarily during periods of high renewable energy production.

4.2.3. Advanced Economic Considerations: Maximizing ROI

Beyond energy efficiency, our approach also emphasizes economic viability. By incorporating the COE, hardware, and the fluctuating price of cryptocurrencies, the algorithm seeks to maximize ROI for mining operations. It not only reduces operational costs by maximizing the use of cheaper renewable energy but also identifies the optimal time frames for cryptocurrency mining to coincide with favorable energy and cryptocurrency market conditions.

4.2.4. Comparison with Existing Algorithms

When compared with the existing energy optimization algorithms, our approach demonstrates several clear benefits:

- **Comprehensive Load Management:** Traditional algorithms often focus on either renewable energy integration or general energy cost minimization. In contrast, our approach specifically addresses the complexities of high-energy-consumption operations like cryptocurrency mining, which are often ignored in conventional algorithms.
- **Dual Optimization Focus:** Many existing algorithms prioritize either technical performance (such as system efficiency) or financial metrics (such as cost minimization). Our algorithm offers a dual focus, optimizing both technical performance (ensuring maximum use of renewable resources) and economic benefits (through cost-effective energy distribution and increased cryptocurrency mining profitability).
- **Scalability:** The proposed method is highly scalable, allowing it to be adapted for various energy systems and mining setups, ranging from small-scale hybrid systems to large-scale industrial applications.

4.2.5. Benefits of the Proposed Algorithm

- **Improved Energy Utilization:** By dynamically managing the energy supply in response to real-time renewable energy generation and cryptocurrency mining demand, the proposed algorithm ensures more efficient energy use, reducing unnecessary energy storage or fossil fuel consumption.
- **Cost Savings and Reduced Emissions:** Increased reliance on renewable energy resources not only results in significant cost savings for mining operations but also contributes to lower greenhouse gas emissions by reducing dependence on diesel generators or other fossil-fuel-based energy sources.
- **Enhanced Flexibility:** The flexibility of the proposed algorithm, which can adapt to market changes in energy prices and cryptocurrency values, ensures that the system remains profitable even under variable conditions, providing an additional layer of robustness compared to traditional models.

In conclusion, the proposed optimization algorithm offers a novel and robust approach to managing hybrid energy systems in the presence of cryptocurrency mining operations. Its ability to integrate real-time data, optimize both technical and economic performance, and offer scalability makes it a superior alternative to existing methods.

5. Case Study

5.1. System Description

The case study focuses on a hybrid renewable energy system designed to meet the energy demands of both a local urban load and a cryptocurrency mining operation. This system is situated in Kashan, Iran, which is characterized by a semi-arid climate with substantial solar irradiance and moderate wind speeds throughout the year. The key components of the energy system include PV panels, wind turbines, a diesel generator, and

a battery storage system, all of which are coordinated to provide an uninterrupted energy supply to both the urban grid and the mining farm.

5.1.1. Location and Climate Data

Kashan (with a longitude of $51^{\circ}24'$ E, latitude of $33^{\circ}59'$ N, and an elevation of 950 m above sea level) offers favorable conditions for renewable energy production, particularly solar energy. The region experiences an average solar irradiance of around $5.5 \text{ kWh/m}^2/\text{day}$ and moderate wind speeds that fluctuate between 3 and 6 m/s, making it an ideal location for a hybrid renewable energy system. These environmental conditions allow for a high penetration of renewable energy into the system, reducing reliance on diesel generators for backup power. Figure 4 illustrates the renewable resource data of the Kashan region, highlighting the annual average solar and wind potentials used for simulation analysis.

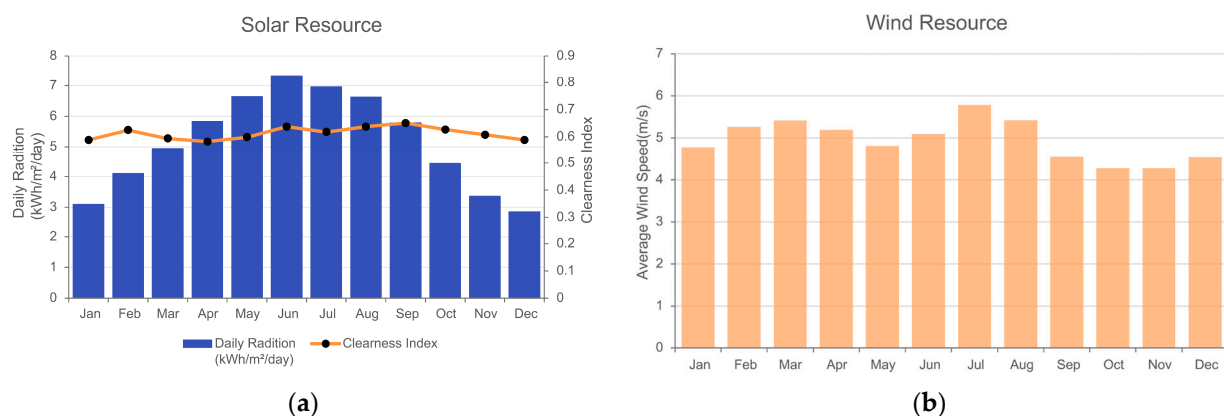


Figure 4. Renewable Resource Data of the Kashan region; (a) Solar Resource—Scaled Annual Average($\text{kWh/m}^2/\text{day}$): 5.16; (b) Wind Resource—Scaled Annual Average(m/s): 4.95.

5.1.2. Components of the Hybrid Energy System

- **PV Panels:** The system employs high-efficiency solar panels as the primary source of renewable energy. These panels are positioned to maximize solar exposure throughout the day and contribute significantly to the overall energy mix, especially during daylight hours.
- **Wind Turbines:** To complement the solar energy system, wind turbines are integrated into the system to harness the region's wind energy, particularly during nighttime and low solar irradiance periods. The combination of solar and wind energy ensures a more balanced and continuous renewable energy supply.
- **Diesel Generator:** A diesel generator serves as a backup power source, ensuring energy availability during periods of low renewable energy generation (e.g., during prolonged cloudy or windless days). The generator is configured to operate only when necessary, minimizing fuel consumption and operational costs.
- **Battery Storage System:** A battery energy storage system is installed to store excess energy generated during peak renewable energy production periods. This stored energy is used to power the mining operation and the local grid during times of low renewable generation or high demand, optimizing system efficiency and reliability.
- **Power Converter:** The system includes a power converter to manage the alternating current (AC) and direct current (DC) flows between the various components, ensuring seamless integration between renewable sources, the diesel generator, and the local loads.

5.1.3. Load Profile

- **Urban Load:** The urban load primarily consists of residential and commercial energy demands, which are relatively stable throughout the day but experience peaks in the morning and evening. This load requires a continuous and reliable energy supply.
- **Cryptocurrency Mining Cooling Loads:** The cryptocurrency mining farm, which runs equipment for digital coin mining, creates a high, continuous, and energy-demanding load. The mining hardware, along with its cooling systems, operates 24/7, requiring a stable power supply to ensure profitability. While energy consumption levels can vary depending on the specific system configurations and operational approaches, they remain consistently substantial, making up a significant portion of the total system's energy demand.

5.1.4. System Configuration and Optimization Objectives

The hybrid system is designed to optimize the balance between renewable energy production, storage, and backup generation to minimize energy costs while ensuring operational stability for both the urban load and the mining operation. The optimization algorithm plays a crucial role in coordinating the energy resources to achieve the following objectives:

- **Maximization of Renewable Energy Utilization:** The system aims to utilize as much energy from renewable sources as possible to minimize the dependence on the diesel generator, thus reducing fuel costs and carbon emissions.
- **Minimization of Operating Costs:** By prioritizing renewable energy and optimizing the operational schedule of the diesel generator and battery system, the system reduces operational and maintenance costs, especially in the context of cryptocurrency mining, which benefits from lower energy costs.
- **Reliability and Stability:** The system is configured to ensure a reliable power supply, particularly to the mining farm, where even short disruptions in power can lead to significant financial losses.

This case study demonstrates how the proposed hybrid system can efficiently manage both urban energy needs and the high-intensity demands of cryptocurrency mining, with a focus on economic and environmental sustainability.

5.2. Scenario Analysis

In this section, we present a detailed analysis of various operational scenarios for the hybrid renewable energy system described in the previous section. The aim of this scenario analysis is to evaluate the system's performance under different conditions, focusing on the impact of renewable energy generation, fuel prices, and the varying energy demands of cryptocurrency mining and urban loads. The proposed optimization algorithm is applied to each scenario to determine the optimal configuration and operational strategy that maximizes both energy efficiency and ROI.

The following scenarios were considered:

5.2.1. Scenario 1: Base Case—Optimized System with Local Fuel Prices

This scenario evaluates the performance of the hybrid energy system under typical local conditions. It assumes the following:

- The use of locally subsidized diesel fuel prices.
- Renewable energy generation based on historical solar irradiance and wind speed data for Kashan.
- Continuous operation of the cryptocurrency mining farm with a steady load.

Key Observations:

- Non-renewable energy sources (diesel generators) supply a significant portion of the total energy demand.
- The system achieves a balance between cost-effectiveness and energy reliability, with a favorable ROI for mining loads.

Figure 5 shows the concept of Scenario 1.

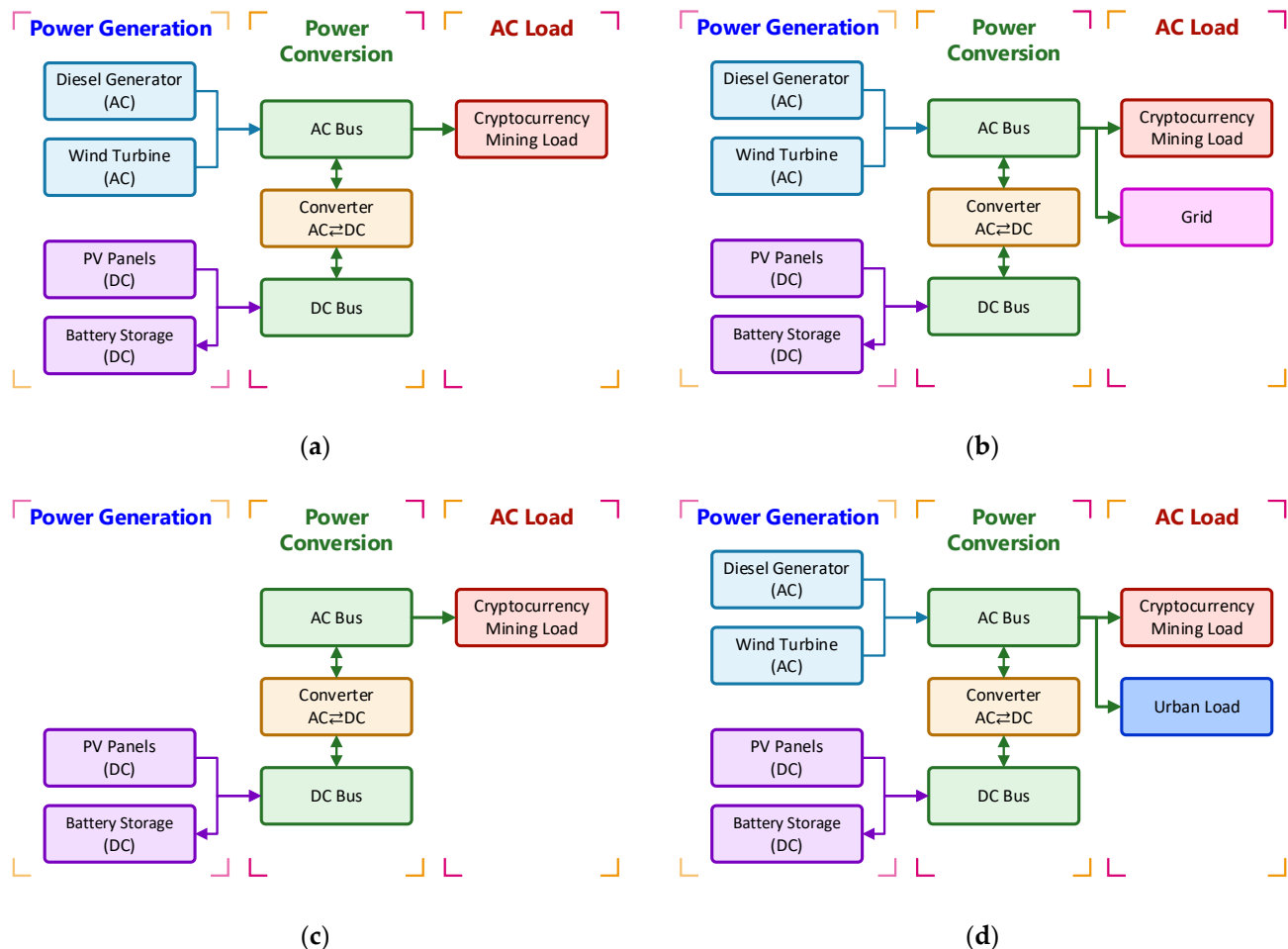


Figure 5. The concept of Scenarios; (a) Scenario 1—Local Fuel Prices; Scenario 2—Global Fuel Price; Scenario 6—Model Validation and Climate-Based Adjustments; Scenario 7—Flexible Mining; Scenario 8—Seasonal Configuration; (b) Scenario 3—Energy Sales; (c) Scenario 4—PV-Only; (d) Scenario 5—Hybrid Load.

5.2.2. Scenario 2: High Fuel Prices—Optimized System with Global Diesel Price

This scenario investigates the effect of global diesel fuel prices, simulating a situation where local subsidies are reduced, or fuel costs increase due to market volatility. The goal is to assess the economic feasibility of the system under these conditions.

Key Observations:

- The increase in fuel prices significantly impacts the operating costs of the system, particularly during periods of low renewable energy production.
- The optimization algorithm shifts to prioritize renewable energy sources even further, reducing the diesel generator's runtime to mitigate high fuel costs.
- Battery storage becomes even more critical in ensuring the system's cost-efficiency, as it helps reduce dependency on expensive diesel power.
- The overall system costs increase, and the profitability of cryptocurrency mining decreases.

Figure 5 shows the concept of Scenario 2.

5.2.3. Scenario 3: High Mining Energy Sales to the National Grid During Peak Summer Months

This scenario evaluates the financial and operational impacts of selling energy produced by the microgrid to the national grid during peak summer months (July and August) when electricity demand is highest. The aim is to support the national grid during periods of frequent power outages while analyzing how this affects the profitability and ROI of the hybrid energy system. The diesel generator operates at the government-subsidized rate of 0.095 USD (6000 IR Rial) per liter.

Scenario Details:

Scenario A: Energy sales to the national grid in July, while the microgrid supplies the cryptocurrency mining farm for the remaining 11 months.

Scenario B: Energy sales to the national grid during both July and August, with the microgrid supplying the mining farm for the remaining 10 months.

Scenario C: Split operation during July, where energy is sold to the national grid from 6 AM to 10 PM, and the microgrid supplies the mining farm from 10 PM to 6 AM. For the rest of the year, the microgrid powers the mining farm.

The microgrid sells energy at a guaranteed rate of 0.058 USD/kWh, based on the 2022 tariff for renewable energy systems. The goal is to assess the economic impact of diverting energy from the mining farm to the national grid during the hottest months when electricity demand surges.

Among the three scenarios, Scenario C offers the best balance between maintaining mining operations and maximizing revenue from energy sales. By selling energy during peak hours and mining during off-peak hours, the system can increase its overall revenue while continuing to benefit from cryptocurrency mining. Both Scenario A and Scenario B result in revenue from energy sales but at the cost of lost mining profits during downtime.

Figure 5 shows the concept of Scenario 3.

5.2.4. Scenario 4: PV—Only Microgrid with Battery Storage

This scenario focuses on a microgrid powered solely by PV panels, supported by battery storage and a power converter. The objective is to evaluate the system's cost, revenue, and payback period, with a 5-year project lifespan and a minimum daily Income of \$50.67.

Key Observations:

- The system operates without fossil fuels, resulting in zero emissions, but its performance is dependent on consistent sunlight, with limited energy storage for cloudy days.

Figure 5 shows the concept of Scenario 4.

5.2.5. Scenario 5: Hybrid Load—Combined Urban and Mining Load Optimization

This scenario explores the simultaneous optimization of both urban and mining loads. It evaluates the system's ability to manage the distinct energy profiles of a local urban area and a cryptocurrency mining operation, each with its own energy demands and consumption patterns.

Key Observations:

- The hybrid load scenario demonstrates the system's flexibility in meeting both urban and mining energy needs through dynamic resource allocation.
- The optimization algorithm effectively balances the intermittent supply of energy with the fluctuating urban load while stabilizing the constant energy demand of the mining farm.

In summary, the scenario analysis highlights the adaptability and robustness of the proposed hybrid energy system. The system performs optimally across various conditions, maintaining profitability and energy reliability while maximizing the use of renewable energy sources. The dynamic optimization approach ensures that the system can handle fluctuations in both energy demand and supply, particularly in the context of high-consumption cryptocurrency mining operations.

Figure 5 shows the concept of Scenario 5.

5.2.6. Scenario 6: Model Validation and Parametric Adjustments for Diverse Climates

This scenario evaluates the applicability of the proposed energy system model across multiple geographic locations with varying climatic conditions. Four sites with distinct solar and wind resource profiles were selected: Palm Springs, USA; Skopun, Faroe Islands; Comodoro Rivadavia, Argentina; and Trondheim, Norway. Figures 6 and 7 illustrate the solar irradiance and wind speed distributions at these locations.

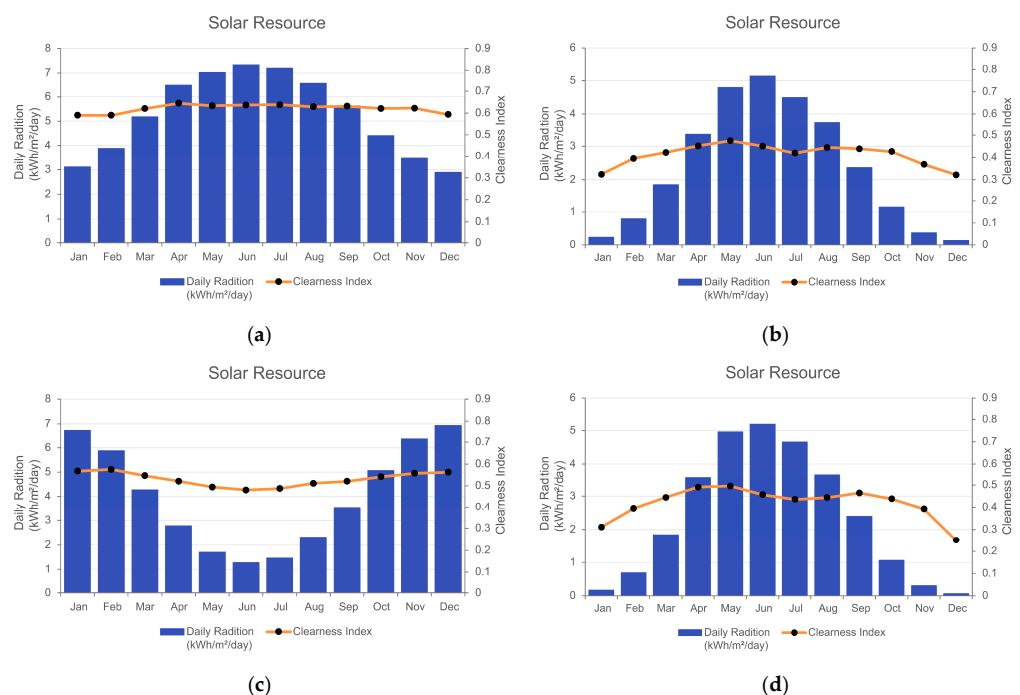


Figure 6. Global Comparison of Solar Irradiance Profiles at Four Representative Sites; (a) 3400 Kirk Douglas Way, Palm Springs, CA 92262, USA; Scaled Annual Average (kWh/m²/day): 5.26; (b) V3VQ+37 Skopun, Faroe Islands; Scaled Annual Average (kWh/m²/day): 2.37; (c) Francina Girardez 547, U9000DSA Comodoro Rivadavia, Chubut, Argentina; Scaled Annual Average (kWh/m²/day): 4.02; (d) Kongens gate 7013, Trondheim, Norway; Scaled Annual Average (kWh/m²/day): 2.38.

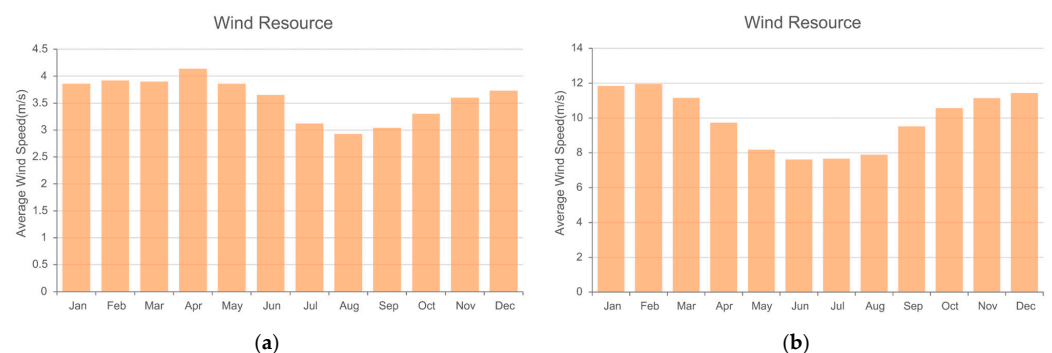


Figure 7. Cont.

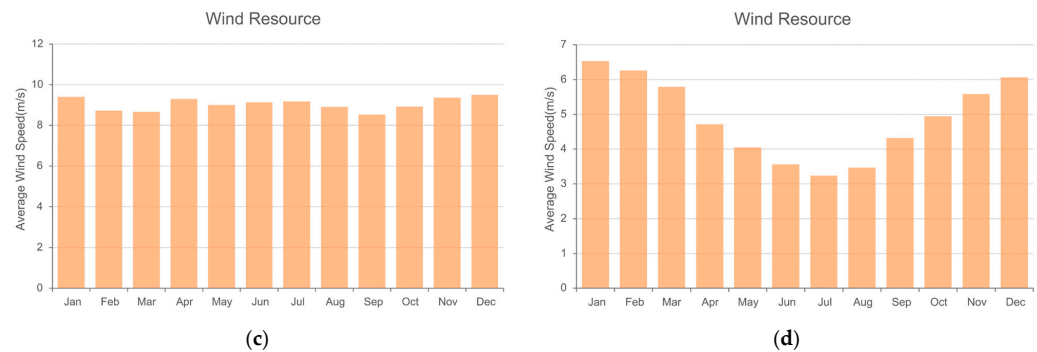


Figure 7. Wind Speed Distribution at Four Geographically Diverse Locations; (a) 3400 Kirk Douglas Way, Palm Springs, CA 92262, USA; Scaled Annual Average(m/s): 3.59; (b) V3VQ+37 Skopun, Faroe Islands; Scaled Annual Average(m/s): 9.89; (c) Francina Girardez 547, U9000DSA Comodoro Rivadavia, Chubut, Argentina; Scaled Annual Average(m/s): 9.05; (d) Kongens gate 7013, Trondheim, Norway; Scaled Annual Average(m/s): 4.88.

Key considerations:

- Adjustments to PV capacity factors were made to reflect differences in solar irradiance between high-latitude and low-latitude regions.
- Wind turbine output estimations incorporated local wind speed variations using standard power curves to assess impacts on diesel fuel consumption.
- These parametric adjustments enable the model to better represent diverse environmental conditions, enhancing its generalizability.

In summary, the scenario underscores the adaptability and robustness of the proposed hybrid energy system. It demonstrates consistent performance across diverse climatic conditions, ensuring energy reliability and economic viability while maximizing the integration of renewable energy sources. The dynamic optimization framework equips the system to effectively manage fluctuations in both energy demand and supply, including in high-demand applications such as cryptocurrency mining operations.

Figure 5 shows the concept of Scenario 6.

5.2.7. Scenario 7: Flexible Crypto Mining and System Resilience Under Renewable and Fuel Price Uncertainty

This scenario examines the capacity of the hybrid energy system to maintain operational and economic performance under highly uncertain renewable availability and fuel pricing conditions. A key strategy in enhancing system resilience involves integrating flexibility into the cryptocurrency mining load, allowing the system to dynamically adjust consumption during periods of limited generation or high operating costs. This load-side flexibility is particularly important in isolated or resource-constrained systems where renewable intermittency and fuel volatility can significantly impact system stability [93].

To capture this behavior, the mining and cooling demand was modeled as a deferrable load within HOMER Pro, reflecting its schedulable and time-shiftable nature. This modeling approach enables the system to temporarily postpone mining operations when supply is insufficient or cost-prohibitive, and to resume them when favorable conditions return. Table 10 outlines the key parameters used to represent this load configuration, assuming a setup of 20 cryptocurrency mining units operating continuously.

Table 10. Deferrable Load Parameters for Mining and Cooling Operations.

Parameter	Value	Description
Peak Load	91.94 kW	Maximum combined power demand of miners and cooling systems
Scaled Average	2206.56 kWh/day	Total daily energy requirement based on continuous operation
Minimum Load Ratio	5%	Minimum allowable operating level
Storage Capacity	1103.28 kWh	Equivalent to 50% of daily load; enables up to half-day interruption

The Load Following dispatch strategy was selected in HOMER Pro to simulate realistic system behavior. In this strategy, renewable and stored energy sources are used to supply the load as a priority, while generators operate only when necessary to meet the remaining demand. As mining constitutes the only significant load in this scenario, the dispatch algorithm dynamically determines when and how much mining activity can be sustained based on system resource availability.

In addition to HOMER Pro's built-in operating reserve mechanisms—which help buffer against sudden drops in renewable output or spikes in load—this scenario introduces a structured sensitivity analysis to further explore the system's performance under diverse and uncertain operating conditions.

HOMER Pro allows for sensitivity analysis by enabling multiple values for selected input variables. It performs optimization for each value or combination of values, making it possible to assess how changes in external conditions affect system performance and costs. This functionality is especially valuable in resilience studies, where uncertainty in fuel pricing and renewable resources plays a critical role in long-term planning and system design.

The following variables, listed in Table 11, were selected as sensitivity inputs to evaluate the impact of extreme scenarios related to resource and cost variability:

Table 11. Sensitivity Input Variables for Resilience-Oriented Analysis.

Sensitivity Input	Values
Diesel Fuel Price (\$/L)	0.6, 0.95, 1.0, 1.8
Solar Scaled Average (kWh/m ² /day)	3, 4, 5, 6, 7
Wind Scaled Average (m/s)	3, 4, 5, 6

These sensitivity parameters were analyzed both individually and in combination, forming a multidimensional framework to evaluate how the system adapts to compound disturbances such as low renewable availability coinciding with elevated fuel costs.

In summary, Scenario 7 highlights the critical role of load flexibility in maintaining resilience and economic efficiency in renewable-based energy systems. By leveraging the schedulability of the mining load and incorporating extensive sensitivity modeling, the system demonstrates an enhanced ability to respond to uncertainty while maintaining service reliability and minimizing operational costs.

In addition to enhancing resilience during fuel price volatility and renewable scarcity, the deferrable mining load also provides an effective mechanism for managing energy surpluses. During periods of exceptionally high solar irradiation or wind availability, mining operations can be scheduled to coincide with excess generation, thereby acting as a controllable energy sink. This not only prevents renewable curtailment but also improves overall system efficiency and renewable utilization, further supporting both economic and environmental objectives.

Figure 5 shows the concept of Scenario 7.

5.2.8. Scenario 8: Seasonal Energy Configuration for Enhanced Mining and System Performance

This scenario investigates the seasonal adaptability of the hybrid energy system in response to variations in cryptocurrency mining demand and renewable resource availability throughout the year. In line with the methodology proposed by Du et al. [94], the objective is to explore how adjusting the internal configuration of energy supply and load characteristics can improve overall system coordination, minimize operational costs, and enhance mining revenue under seasonally changing conditions.

Table 12 outlines the monthly average electricity consumption, which reflects increased mining and cooling demand during warmer months. This seasonal rise is not only driven by higher computational and cooling requirements but also by the greater availability of renewable energy resources during these months. Specifically, enhanced solar irradiance and wind speeds in summer months lead to increased renewable generation potential, thus justifying higher mining activity. Conversely, during the colder months, reduced solar exposure and lower wind speeds significantly constrain renewable generation. As a result, a lower mining load is assumed in these periods to align with limited resource availability. This seasonal adjustment was made in accordance with measured solar radiation and wind speed coefficients for the project location.

Table 12. Monthly Mining Load Profile (kWh).

Month	Daily Radiation (kWh/m ² /day)	Average Wind Speed (m/s)	K (Renewable Supply Coefficient)	Load (kWh)
Jan	3.090	4.770	0.6	$0.6 \times 91.94 = 55.17$
Feb	4.100	5.260	0.75	68.96
Mar	4.940	5.410	0.8	73.56
Apr	5.830	5.190	0.85	78.15
May	6.640	4.800	0.9	82.75
Jun	7.340	5.090	1	91.94
Jul	6.960	5.780	1	91.94
Aug	6.620	5.420	0.95	87.35
Sep	5.780	4.550	0.8	73.56
Oct	4.430	4.280	0.7	64.36
Nov	3.360	4.280	0.6	55.17
Dec	2.850	4.540	0.6	55.17

Note: Fuel costs were calculated based on global diesel prices. K represents the share of total power generation provided by renewable sources, specifically solar and wind energy. It is calculated based on the actual output of PV panels and wind turbines over the simulation period and reflects the relative contribution of renewables to the overall energy mix. ($K \times \text{Total power demand of 20 mining units} = \text{Load}$).

In parallel, high-resolution monthly solar and wind resource profiles were incorporated to simulate the seasonal fluctuation in renewable energy production. While HOMER Pro does not allow direct seasonal reconfiguration of installed generation capacities or binary investment decisions (as enabled by MILP or Scenario-Based Benders Decomposition approaches), it dynamically adjusts system dispatch based on resource availability and demand conditions.

HOMER Pro uses a rule-based dispatch algorithm, notably the Combined Dispatch (CD) strategy, which enables the generator to operate in coordination with the battery storage system to minimize total operating costs rather than merely following the load. Unlike Load Following, CD allows the generator to run at optimal efficiency points even when the load is low, by charging batteries with surplus capacity, thereby reducing fuel consumption over time. This method results in a more globally optimized system behavior, especially when dealing with fluctuating renewable inputs and load demands.

Although the software lacks the capability to execute complex reconfiguration strategies such as MILP-based seasonal topology switching, it compensates by integrating temporal profiles that influence system behavior indirectly. This includes the following:

- Utilizing generator capacity more strategically through CD logic to charge storage when beneficial;
- Responding dynamically to seasonal shifts in mining load;
- Leveraging internal cost minimization routines to favor renewable dispatch when economically and technically feasible.

Thus, HOMER Pro inherently captures a simplified form of seasonal coordination through its dispatch logic and temporal input variability. By enabling a more cost-optimal generator operation regime, CD provides a closer approximation to the flexible and adaptive planning envisioned in advanced scenario-based methodologies. While not equivalent to scenario-based decomposition methods or dynamic investment optimization, the adopted approach delivers a realistic and computationally tractable framework to explore seasonality in system performance.

In summary, Scenario 8 demonstrates how integrating seasonal load and resource data within HOMER Pro's CD environment can achieve a more holistic and efficient energy management strategy, approximating flexible system behavior across the year. For more advanced and granular planning—including discrete seasonal reconfigurations and investment optimization—future studies are encouraged to adopt MILP-capable modeling platforms such as Pyomo, GAMS, or MATLAB, which support binary variables, uncertainty modeling, and scenario decomposition techniques as discussed in [94].

Figure 5 shows the concept of Scenario 8.

5.3. Simulation Results

The following section presents the results of the simulations conducted for the hybrid renewable energy system described in the case study using the scenarios outlined earlier. The simulations were performed using the HOMER Pro software, which allowed for a comprehensive evaluation of the system's performance under various conditions. The key performance indicators (KPIs) considered include total energy production, renewable energy contribution, fuel consumption, system costs, and ROI. Below is an in-depth analysis of the simulation results for each scenario.

5.3.1. Scenario 1: Base Case—Optimized System with Local Fuel Prices

- **Total Energy Production:** The system produced 805 MWh/year, with renewable energy sources contributing 0% of the total demand.
- **Renewable Energy Utilization:** The system achieved a high utilization of non-renewable energy.
- **Fuel Consumption:** The diesel generator consumed 322,291 L of fuel per year.
- **Cost and ROI:** The total annual operational cost, including fuel and maintenance for the energy system, amounted to \$14,920. The ROI was favorable, with a payback period of approximately 1 year due to the local fuel prices.
- **Emissions:** The system's extensive use of diesel systems results in substantial greenhouse gas emissions of approximately 854,872 kg per year, causing significant environmental damage.

Tables 13–15 provide an integrated view of Scenario 1 under local fuel prices. Table 13 presents the cost and revenue distribution across different miner types under subsidized fuel, highlighting the economic performance. Table 14 reports the associated microgrid emissions for the Bitmain Antminer AL1 Pro (16.6 Th) load, emphasizing the environmental impact. Table 15 then describes the corresponding microgrid configurations, connecting the economic and emission outcomes to the system setup employed.

Table 13. Cost-Revenue Breakdown by Miner Type with Subsidized Fuel.

Miner Model	Best Price	Top Minalbe Coins Based on Profit	Income/Day	NPC(Miners) for 20 Qty	O&M Miners	The Cost of Renting, Labor, Internet Bandwidth	NPC (Energy System—Local Fuel Cost)	Total Power for Miners Consumption (kW)	Cooling Power Required (kW) ($\eta = 0.7$, COP = 4)	Total Power Consumption (kW)	Total Income (365 Days—1 Year)	Total Income with USA Inflation Rate for 5 Years	Total NPC	Income 1st Year	Income 2nd Year	Income 3rd Year	Income 4th Year	Income 5th Year	IRR
Bitmain Antminer AL1 Pro (16.6 Th)	\$9990	ALPH	\$50.67	\$199,800	\$39,960	\$20,000	\$62,534	74.6	17.3445	91.9445	\$369,891	\$1,619,298	\$322,294	\$353,624.28	\$338,072.93	\$323,205.48	\$308,991.85	\$295,403.30	103%
Bitmain Antminer AL1 (15.6 Th)	\$9299	ALPH	\$47.62	\$185,980	\$37,196	\$20,000	\$60,044	70.2	16.3215	86.5215	\$347,626	\$1,521,827	\$303,220	\$332,338.43	\$317,723.17	\$303,750.64	\$290,392.58	\$277,621.97	103%
IceRiver AL3 (15.6 Th)	\$9450	ALPH	\$45.79	\$189,000	\$37,800	\$20,000	\$60,044	70	16.275	86.275	\$334,267	\$1,463,344	\$306,844	\$319,566.92	\$305,513.31	\$292,077.73	\$279,233.01	\$266,953.17	97%
Bitmain Antminer S21e XP Hyd 3U	\$19,705	BTC	\$40.43	\$394,100	\$78,820	\$20,000	\$186,992	223.6	51.987	275.587	\$295,139	\$1,292,051	\$679,912	\$282,159.66	\$269,751.11	\$257,888.25	\$246,547.08	\$235,704.67	27%
Bitmain Antminer L9 (16 Gh)	\$6450	DOGE& LTC	\$25.59	\$129,000	\$25,800	\$20,000	\$56,151	67.2	15.624	82.824	\$186,807	\$817,798	\$230,951	\$178,591.78	\$170,737.84	\$163,229.29	\$156,050.95	\$149,188.29	68%
Bitmain Antminer L9 (17 Gh)	\$6630	DOGE& LTC	\$24.92	\$132,600	\$26,520	\$20,000	\$55,075	65.2	15.159	80.359	\$181,916	\$796,386	\$234,195	\$173,915.87	\$166,267.56	\$158,955.60	\$151,965.20	\$145,282.22	65%
ElphaPex DG 1+	\$7521	DOGE& LTC	\$22.39	\$150,420	\$30,084	\$20,000	\$67,399	78.4	18.228	96.628	\$163,447	\$715,533	\$267,903	\$156,259.08	\$149,387.27	\$142,817.66	\$136,536.96	\$130,532.46	47%
Bitmain Antminer S21 XP Hyd (473 Th)	\$9100	BTC	\$22.24	\$182,000	\$36,400	\$20,000	\$120,030	113.52	26.3934	139.9134	\$162,352	\$710,740	\$358,430	\$155,212.24	\$148,386.46	\$141,860.86	\$135,622.24	\$129,657.97	29%
Bitmain Antminer KS5 Pro (21 Th)	\$2650	KAS	\$17.66	\$53,000	\$10,600	\$20,000	\$54,355	63	14.6475	77.6475	\$128,918	\$564,373	\$137,955	\$123,248.57	\$117,828.46	\$112,646.71	\$107,692.84	\$102,956.82	81%
Bitmain Antminer KS5 (20 Th)	\$2180	KAS	\$16.30	\$43,600	\$8720	\$20,000	\$50,255	60	13.95	73.95	\$118,990	\$520,911	\$122,575	\$113,757.17	\$108,754.46	\$103,971.76	\$99,399.39	\$95,028.10	85%
ElphaPex DG 1	\$4940	DOGE& LTC	\$16.08	\$98,800	\$19,760	\$20,000	\$57,830	68.4	15.903	84.303	\$117,384	\$513,880	\$196,390	\$112,221.80	\$107,286.61	\$102,568.46	\$98,057.80	\$93,745.51	46%

Note: Annual income refers to the total revenue generated over a 365-day period. The present value of this income over 5 years was calculated using the inflation rate of approximately 4.6% (2024), accounting for the decrease in purchasing power. The total discounted income is the sum of each year’s income adjusted by the factor $\frac{1}{(1+0.046)^n}$, where n is the year number from 1 to 5. This method underpins the IRR calculations reported in the table.

Table 14. Microgrid Emissions (Local Fuel Prices, AL1 Pro Load).

Emission Species	Config 1 *	Config 2	Config 3	Config 4	Config 5	Config 6	Config 7	Config 8
Carbon Dioxide (kg/year)	852,688	852,340	849,183	849,143	1,385,746	1,385,744	1,382,240	1,381,243
Carbon Monoxide (kg/year)	12.9	12.8	12.8	12.8	21	21	20.9	20.9
Sulfur Dioxide (kg/year)	9.67	9.66	9.63	9.63	15.7	15.7	15.7	15.7
Particulate Matter (kg/year)	31.6	31.6	31.5	31.5	51.3	51.3	51.2	51.2
Nitrogen Oxides (kg/year)	2111	2110	2102	2102	3431	3431	3422	3420
Unburned Hydrocarbons (kg/year)	19.3	19.3	19.3	19.3	31.4	31.4	31.3	31.3
Total Emissions (kg/year)	854,872	854,523	851,358	851,318	1,389,296	1,389,294	1,385,781	1,384,782

* Note: “Config” is used as an abbreviation for “Configuration.”

Table 15. Microgrid Configurations—Scenario 1 (Local Fuel Prices, AL1 Pro Load).

Configuration Number	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/year)
1	-	-	✓*	✓	✓	\$62,533.88	0.01588658	854,872
2	✓ 0.502	-	✓	✓	✓	\$63,039.56	0.01601505	854,523
3	-	✓ 10	✓	✓	✓	\$67,453.81	0.01713648	851,358
4	✓ 0.0578	✓ 10	✓	✓	✓	\$67,487.29	0.01714498	851,318
5	-	-	✓	-	-	\$117,103.30	0.02974981	1,389,296
6	✓ 0.00182	-	✓	-	✓	\$117,239.10	0.02978431	1,389,294
7	-	✓ 10	✓	-	-	\$122,189.80	0.03104202	1,385,781
8	✓ 2.94	✓ 10	✓	-	✓	\$124,521.60	0.03163441	1,384,782

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

Figure 8 depicts the relationship between the renewable energy share and the internal rate of return (IRR) for scenario 1: base case—optimized system with local fuel prices, with a trendline highlighting the overall correlation and scatter points representing different microgrid configurations.

As can be seen, the scatter plot between the renewable energy share and the IRR reveals no strong linear correlation. This suggests that higher renewable shares do not necessarily result in greater economic returns. The main reasons are the high initial capital costs of renewable technologies and their variable output, which may not sufficiently compensate for the investment.

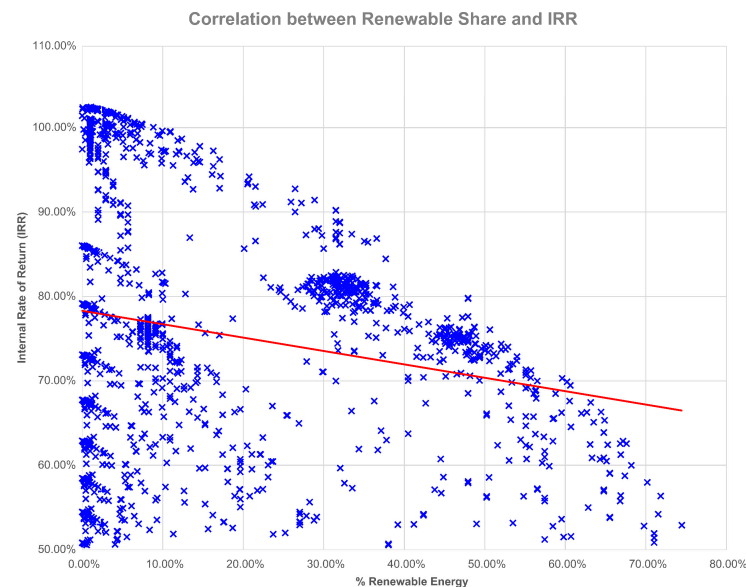


Figure 8. Scatter Plot of Renewable Share vs. IRR. The red line represents the trendline indicating the overall relationship, and the blue markers denote different configurations. %Renew refers to the share of energy produced from renewable sources relative to total installed capacity, and IRR indicates the internal rate of return.

5.3.2. Scenario 2: High Fuel Prices—Optimized System with Global Diesel Price

- **Total Energy Production:** The total energy output remained consistent at 914 MWh/year, but the cost implications of diesel price increases were more pronounced.
- **Renewable Energy Utilization:** The optimization algorithm was adjusted to maximize renewable energy usage, resulting in a higher renewable energy contribution of 37.7%, as the system aimed to reduce fuel consumption.
- **Fuel Consumption:** Diesel consumption decreased to 224,574 L per year as the system minimized generator use to mitigate the impact of higher fuel prices.
- **Cost and ROI:** Despite reducing fuel consumption, the annual operational costs increased to \$68,670 due to the higher fuel prices. The ROI remained positive, though the payback period was extended to 20 years due to the additional fuel expenses.
- **Emissions:** The system reduced carbon emissions by 49% compared to a fully diesel-powered system, contributing significantly to environmental sustainability.

Tables 16–18 illustrate the performance of the system under Scenario 2 with global diesel prices. Table 16 presents the cost and revenue distribution across different miner types, highlighting the economic implications. Table 17 summarizes the corresponding microgrid emissions for the AL1 Pro load, providing insight into the environmental impact. Table 18 then details the microgrid configurations, linking the observed economic and emission outcomes to the system setup applied in this scenario.

Table 16. Cost-Revenue Breakdown by Miner Type with Global Diesel Price.

Miner Model	Best Price	Top Minalbe Coins Based on Profit	Income/Day	NPC(Miners) for 20 Qty	O&M Miners	The Cost of Renting, Labor, Internet Bandwidth	NPC (Energy System—USA Fuel Cost)	Total Power for Miners Consump-tion (kW)	Cooling Power Required (kw) ($\eta = 0.7$, $COP = 4$)	Total Power Consumption (kW)	Total Income (365 Days—1 Year)	Total Income with USA Inflation Rate for 5 Years	Total NPC	Income 1st Year	Income 2nd Year	Income 3rd Year	Income 4th Year	Income 5th Year	IRR
Bitmain Antminer AL1 Pro (16.6 Th)	\$9990	ALPH	\$50.67	\$199,800	\$39,960	\$20,000	\$1,350,975	74.6	17.3445	91.9445	\$369,891	\$1,619,298	\$1,610,735	\$353,624.28	\$338,072.93	\$323,205.48	\$308,991.85	\$295,403.30	0.2%
Bitmain Antminer AL1 (15.6 Th)	\$9299	ALPH	\$47.62	\$185,980	\$37,196	\$20,000	\$1,100,000	70.2	16.3215	86.5215	\$347,626	\$1,521,827	\$1,343,176	\$332,338.43	\$317,723.17	\$303,750.64	\$290,392.58	\$277,621.97	4%
IceRiver AL3	\$9450	ALPH	\$45.79	\$189,000	\$37,800	\$20,000	\$1,100,000	70	16.275	86.275	\$334,267	\$1,463,344	\$1,346,800	\$319,566.92	\$305,513.31	\$292,077.73	\$279,233.01	\$266,953.17	3%
Bitmain Antminer S21e XP Hyd 3U	\$19,705	BTC	\$40.43	\$394,100	\$78,820	\$20,000	\$3,520,000	223.6	51.987	275.587	\$295,139	\$1,292,051	\$4,012,920	\$282,159.66	\$269,751.11	\$257,888.25	\$246,547.08	\$235,704.67	−29%
Bitmain Antminer L9 (16 Gh)	\$6450	DOGE& LTC	\$25.59	\$129,000	\$25,800	\$20,000	\$1,050,000	67.2	15.624	82.824	\$186,807	\$817,798	\$1,224,800	\$178,591.78	\$170,737.84	\$163,229.29	\$156,050.95	\$149,188.29	−12%
Bitmain Antminer L9 (17 Gh)	\$6630	DOGE& LTC	\$24.92	\$132,600	\$26,520	\$20,000	\$1,020,000	65.2	15.159	80.359	\$181,916	\$796,386	\$1,199,120	\$173,915.87	\$166,267.56	\$158,955.60	\$151,965.20	\$145,282.22	−13%
ElphaPex DG 1+	\$7521	DOGE& LTC	\$22.39	\$150,420	\$30,084	\$20,000	\$1,230,000	78.4	18.228	96.628	\$163,447	\$715,533	\$1,430,504	\$156,259.08	\$149,387.27	\$142,817.66	\$136,536.96	\$130,532.46	−20%
Bitmain Antminer S21 XP Hyd (473 Th)	\$9100	BTC	\$22.24	\$182,000	\$36,400	\$20,000	\$1,920,000	113.52	26.3934	139.9134	\$162,352	\$710,740	\$2,158,400	\$155,212.24	\$148,386.46	\$141,860.86	\$135,622.24	\$129,657.97	−29%
Bitmain Antminer KS5 Pro (21 Th)	\$2650	KAS	\$17.66	\$53,000	\$10,600	\$20,000	\$994,366	63	14.6475	77.6475	\$128,918	\$564,373	\$1,077,966	\$123,248.57	\$117,828.46	\$112,646.71	\$107,692.84	\$102,956.82	−19%
Bitmain Antminer KS5 (20 Th)	\$2180	KAS	\$16.30	\$43,600	\$8720	\$20,000	\$936,261	60	13.95	73.95	\$118,990	\$520,911	\$1,008,581	\$113,757.17	\$108,754.46	\$103,971.76	\$99,399.39	\$95,028.10	−19%
ElphaPex DG 1	\$4940	DOGE& LTC	\$16.08	\$98,800	\$19,760	\$20,000	\$1,070,000	68.4	15.903	84.303	\$117,384	\$513,880	\$1,208,560	\$112,221.80	\$107,286.61	\$102,568.46	\$98,057.80	\$93,745.51	−24%

Table 17. Microgrid Emissions (Global Diesel Price, AL1 Pro Load).

Emission Species	Config 1 *	Config 2	Config 3	Config 4	Config 5	Config 6	Config 7	Config 8
Carbon Dioxide (kg/year)	594,158	578,046	791,007	852,688	1,191,290	1,159,044	1,385,746	1,384,818
Carbon Monoxide (kg/year)	8.98	8.74	12	12.9	18	17.5	21	20.9
Sulfur Dioxide (kg/year)	6.74	6.55	8.97	9.67	13.5	13.1	15.7	15.7
Particulate Matter (kg/year)	22	21.4	29.3	31.6	44.1	42.9	51.3	51.3
Nitrogen Oxides (kg/year)	1471	1431	1958	2111	2949	2869	3431	3428
Unburned Hydrocarbons (kg/year)	13.5	13.1	17.9	19.3	27	26.3	31.4	31.4
Total Emissions (kg/year)	595,680	579,527	793,033	854,872	1,194,342	1,162,013	1,389,296	1,388,365

* Note: “Config” is used as an abbreviation for “Configuration.”

Table 18. Microgrid Configurations—Scenario 2 (Global Diesel Price, AL1 Pro Load).

Configuration Number	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/year)
1	-	✓*	✓	✓	✓	\$1,350,975	0.3432118	595,680
		390	90	304	76.9			
2	✓	✓	✓	✓	✓	\$1,352,003	0.343473	579,527
	1.27	440	90	302	77.62			
3	✓	-	✓	✓	✓	\$1,537,458	0.3906226	793,033
	75.56	-	90	209	65.64			
4	-	-	✓	✓	✓	\$1,543,946	0.3922358	854,872
			100	5	1.63			
5	-	✓	✓	-	-	\$2,483,739	0.6309879	1,194,342
		600	200					
6	✓	✓	✓	-	✓	\$2,492,950	0.6333279	1,162,013
	0.92	730	200		0.55			
7	-	-	✓	-	-	\$2,524,618	0.6413732	1,389,296
			200					
8	✓	-	✓	-	✓	\$2,531,970	0.6432409	1,388,365
	11.49		200		0.72			

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

Figure 9 depicts the relationship between the renewable energy share and the internal rate of return (IRR) for scenario 2: high fuel prices—optimized system with global diesel price, with a trendline highlighting the overall correlation and scatter points representing different microgrid configurations.

An observable trend in the scatter plot between renewable energy share and IRR indicates the absence of a strong linear relationship. This suggests that increasing the proportion of installed renewable capacity does not inherently guarantee improved economic performance. The likely explanation lies in the high upfront capital costs and the intermittent nature of renewable generation, which may limit its ability to deliver consistent financial benefits.

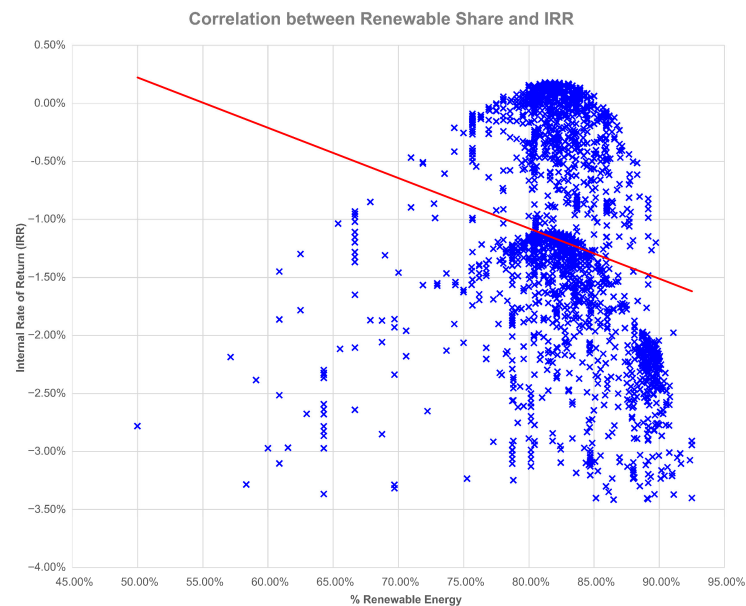


Figure 9. Scatter Plot of Renewable Share vs. IRR. The red line represents the trendline indicating the overall relationship, and the blue markers denote different configurations. %Renew refers to the share of energy produced from renewable sources relative to total installed capacity, and IRR indicates the internal rate of return.

5.3.3. Scenario 3: High Mining Energy Sales to the National Grid During Peak Summer Months

- **Total Energy Production:** The system produced 805 MWh/year, with approximately 1.1% of the energy generated by renewable sources (solar and wind) and the remaining 98.9% provided by the diesel generator.
- **Renewable Energy Utilization:** The system achieved a high utilization of non-renewable energy, with PV panels contributing 0.012% and wind turbines contributing 1.1% of the total energy demand. The remaining energy came from the diesel generator.
- **Fuel Consumption:** The diesel generator consumed 320,000 L of fuel per year.
- **Cost and ROI:** The total annual operational cost, including fuel and maintenance, amounted to \$15,877. The ROI was favorable, with a payback period of more than 1 year.
- **Emissions:** The system's extensive use of diesel systems results in substantial greenhouse gas emissions of approximately 854,872 kg per year, causing significant environmental damage.

Table 19 presents the cost and revenue breakdown under subsidized fuel conditions across different sale scenarios for the Bitmain Antminer AL1 Pro 16.6 Th load.

Table 19. Cost-Revenue Breakdown with Subsidized Fuel under Different Sale Scenarios (Bitmain Antminer AL1 Pro 16.6 Th Load).

Scenario Number	Electricity Generated for Sale to Grid in 1 Year (kWh)	Guaranteed Purchase Price (USD)	Revenue from Sale to Grid in 1 Year (USD) (kWh)	Miner Income (365 Days—1 Year)	Total Income (365 Days—1 Year)	Total Revenue from Electricity Sale to Grid Over Project Lifetime (USD)	Total Farm Downtime in One Year (Hours)	Farm Operation Ratio in a Year (Relative to Full-Time Operation)	Revenue from the Mining Farm over the Project Lifetime (USD)	Total Revenue from the Mining Farm and Grid Electricity Sales over the Project Lifetime (USD)	NPV	Total NPC	IRR
A	68,405	\$0.058	\$3967	\$338,476	\$342,443	\$17,369	744	0.9151	\$1,481,768	\$1,499,137	\$1,176,843	\$322,294	94%
B	136,809	\$0.058	\$7935	\$307,060	\$314,995	\$34,737	1488	0.8301	\$1,344,239	\$1,378,976	\$1,056,682	\$322,294	86%
C	45,603	\$0.058	\$2645	\$348,947	\$351,592	\$11,579	496	0.9434	\$1,527,612	\$1,539,191	\$1,216,897	\$322,294	97%

5.3.4. Scenario 4: PV—Only Microgrid with Battery Storage

- **Total Energy Production:** The system produced 1800 MWh/year, with a total of 100% of the energy generated by the PV.

- **Renewable Energy Utilization:** Renewable energy generation increased to 100%.
- **Fuel Consumption:** Diesel fuel consumption dropped significantly to 0 L per year as the generator was not in this scenario.
- **Cost and ROI:** Energy O&M costs increased to \$502,000, reflecting increased demand for battery energy storage. Mining profits still generate a negative overall ROI.
- **Emissions:** By completely replacing fossil fuels with renewable energy, carbon emissions have been reduced to zero.

Table 20 presents the return on investment (ROI) for a PV-only microgrid with battery storage, highlighting the system's financial performance. Complementing this, Figure 10 illustrates the charge and discharge behavior of the storage system, showing the daily profiles of the state of charge, as well as the charge and discharge power, which underpin the observed ROI outcomes.

Table 20. ROI for Microgrid with PV and Battery.

Equipment	Optimal Capacity	NPC of Microgrid (USD)	COE (USD/kWh)	Emission Rate (kg/Year)	Cost of Mining Equipment (USD) + O&M Miners + The Cost of Renting, Labor, Internet Bandwidth	Total NPC	Revenue from Mining Farm (USD)	NPV (USD)	IRR
PV Storage Converter	1008 (kW) 7361 (kWh) 125 (kW)	\$2,300,000	0.584	0	\$259,760	\$2,559,760	\$1,619,298	−\$940,462	−10%

5.3.5. Scenario 5: Hybrid Load—Combined Urban and Mining Load Optimization

- **Total Energy Production:** The system produced 2865 MWh/year, with renewable energy sources contributing 0% of the total demand.
- **Renewable Energy Utilization:** PV panels and wind turbines did not contribute to the supply, while the diesel generator supplied 100%. The battery storage system played a backup role in managing peak loads and reducing the need for diesel during high-demand periods.
- **Fuel Consumption:** Diesel consumption was optimized at 1,336,469 L per year, as the system prioritized the use of non-renewable energy for both loads because of the fuel subsidy.
- **Cost and ROI:** The energy system's operational costs were \$70,541 per year, and due to subsidized local fuel prices, the system achieved a favorable payback period of 1.4 years.
- **Emissions:** This system's carbon emissions increase by approximately 400% compared to systems with only a mining load connected to it. That is because the urban load is connected to the microgrid.

Figure 11 illustrates the combined urban and mining load over the months, including both the user-defined daily profile and the monthly average of AC primary load served in kW. Building on this load data, Tables 21–23 present the associated economic and environmental performance under subsidized fuel conditions. Table 21 details the cost and revenue breakdown for urban load sale scenarios with the Bitmain Antminer AL1 Pro 16.6 Th load, while Table 22 summarizes the corresponding microgrid emissions under local fuel prices. Table 23 extends the economic analysis to varying combinations of AL1 Pro and urban loads, linking the observed costs and revenues to the system operating conditions.

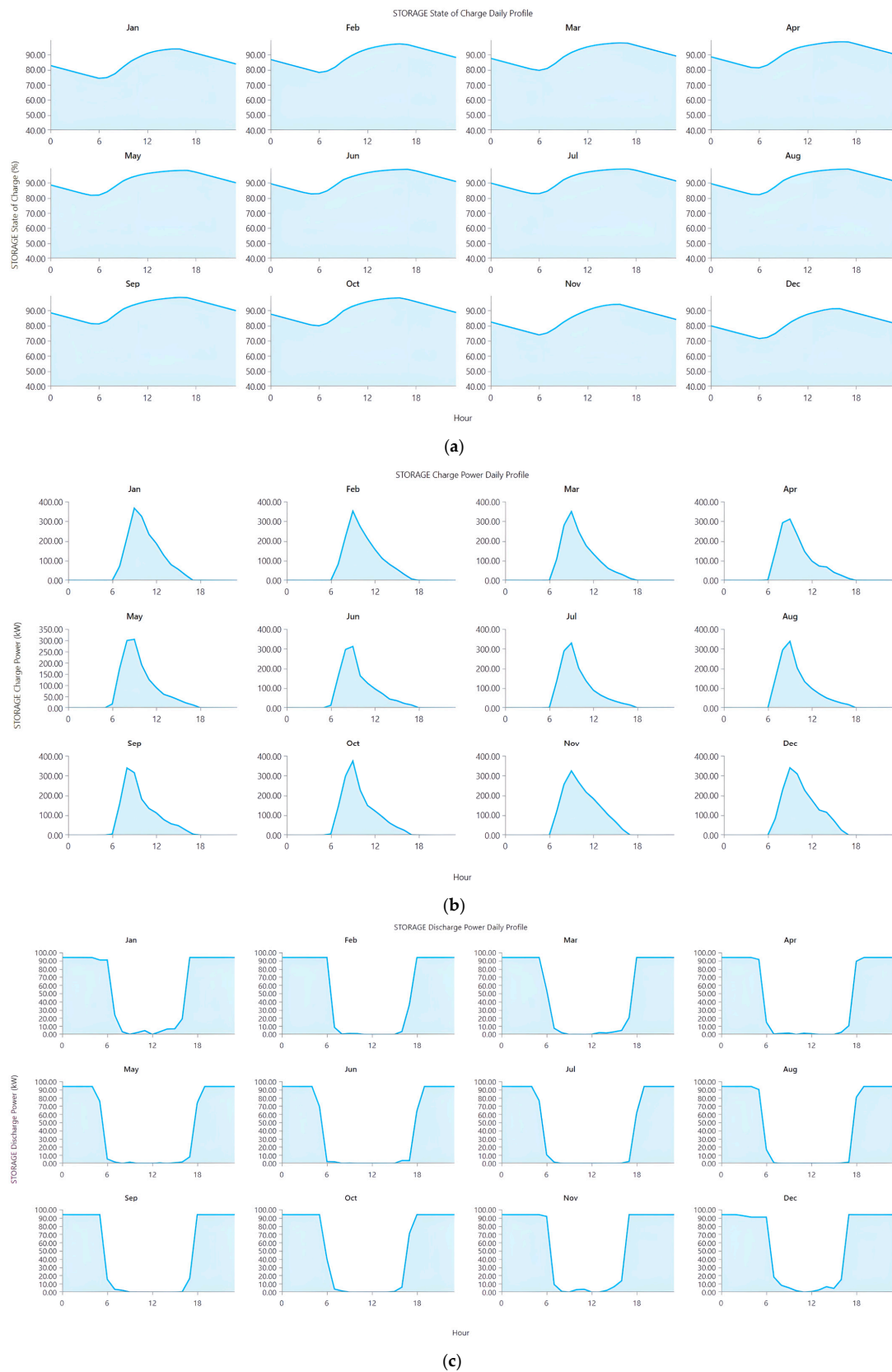


Figure 10. Charge and Discharge Status; (a) Daily Profile of Storage State of Charge; (b) Daily Profile of Storage Charge Power; (c) Daily Profile of Storage Discharge Power.

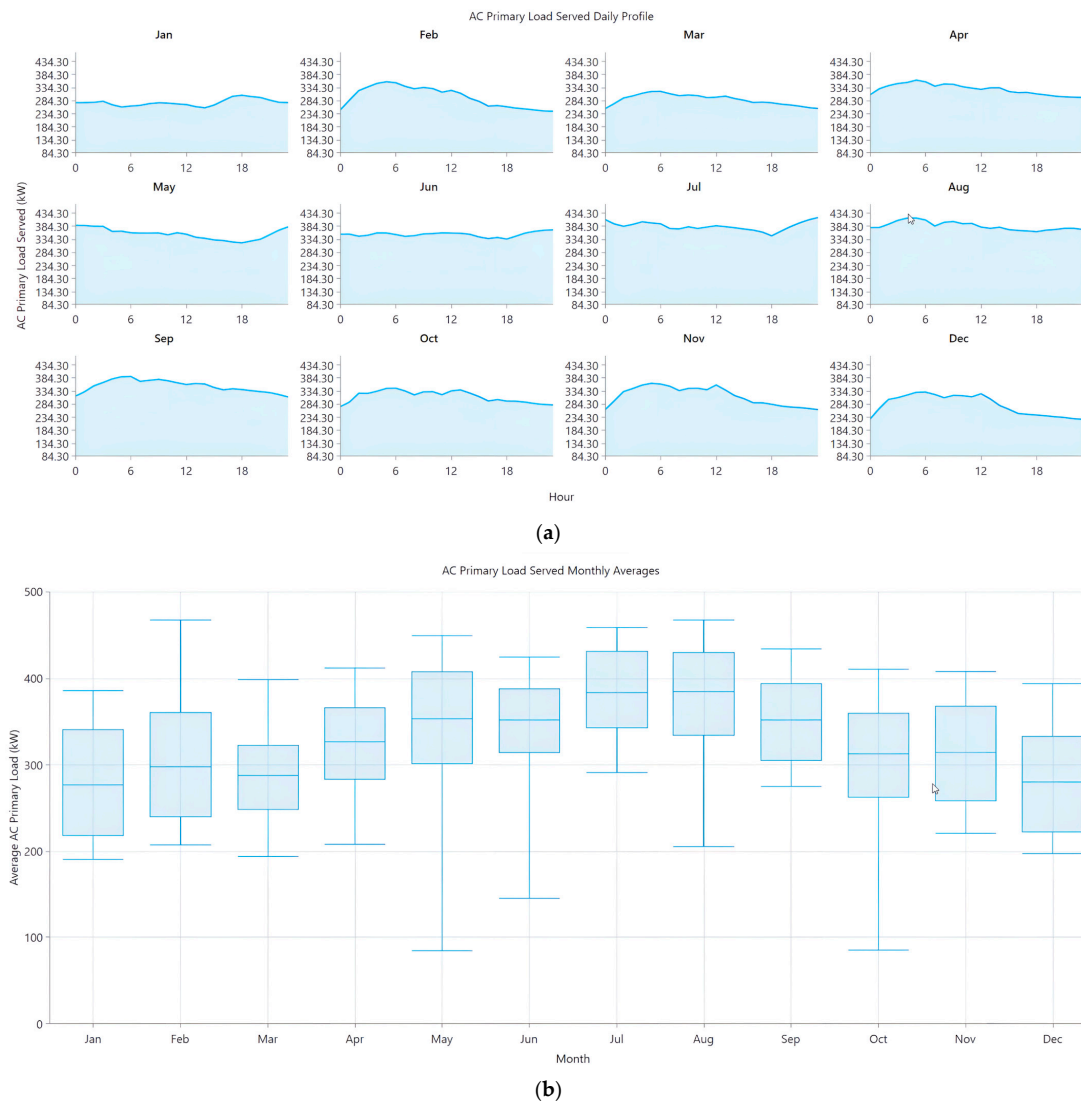


Figure 11. The Combination of Urban and Mining Load data based on the months; (a) AC primary load served (kW)—user-defined daily profile; (b) AC primary load served (kW)—monthly average.

Table 21. Cost-Revenue Breakdown with Subsidized Fuel—Urban Load Sale Scenarios (Bitmain Antminer AL1 Pro 16.6 Th Load).

Electricity Generated for Sale to Grid in 1 Year (kWh)	Guaranteed Purchase Price (USD)	Revenue from Sale to Grid in 1 Year (USD) (kWh)	Miner Income (365 Days—1 Year)	Total Income (365 Days—1 Year)	Total Revenue from Electricity Sale to Grid over Project Lifetime (USD)	Revenue from the Mining Farm over the Project Lifetime (USD)	Total Revenue from the Mining Farm and Grid Electricity Sales over the Project Lifetime (USD)	NPV	Total NPC	IRR
2,060,391	\$0.058	\$119,503	\$369,891	\$489,394	\$523,155	\$1,619,298	\$2,142,453	\$2,689,306.11	\$546,853	77%

Table 22. Microgrid Emissions (Local Fuel Price, AL1 Pro Load).

Emission Species	Config 1 *	Config 2	Config 3	Config 4	Config 5	Config 6	Config 7	Config 8
Carbon Dioxide (kg/year)	3,535,904	3,535,049	3,532,412	3,530,738	3,802,554	3,802,549	3,799,063	3,796,401
Carbon Monoxide (kg/year)	53.5	53.5	53.4	53.4	57.5	57.5	57.4	57.4
Sulfur Dioxide (kg/year)	40.1	40.1	40.1	40	43.1	43.1	43.1	43
Particulate Matter (kg/year)	131	131	131	131	141	141	141	141
Nitrogen Oxides (kg/year)	8754	8752	8745	8741	9414	9414	9405	9399
Unburned Hydrocarbons (kg/year)	80.2	80.2	80.1	80.1	86.2	86.2	86.2	86.1
Total Emissions (kg/year)	3,544,963	3,544,106	3,541,462	3,539,784	3,812,296	3,812,291	3,808,796	3,806,128

* Note: “Config” is used as an abbreviation for “Configuration.”

Table 23. Cost-Revenue Breakdown with Subsidized Fuel—Varying AL1 Pro and Urban Load.

Configuration Number	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/Year)
1	-	-	✓ [*] 450	✓ 69.1	✓ 22.7	\$287,094	0.02050497	3,544,963
2	✓ 1.23	-	✓ 450	✓ 69.1	✓ 22.1	\$287,978	0.02056811	3,544,106
3	-	✓ 10	✓ 450	✓ 68.1	✓ 19.7	\$291,683	0.02083276	3,541,462
4	✓ 2.42	✓ 10	✓ 450	✓ 71.1	✓ 21.6	\$294,324	0.02102135	3,539,784
5	-	-	✓ 500	-	-	\$298,693	0.0213326	3,812,296
6	✓ 0.00649	-	✓ 500	-	✓ 7.17	\$299,382	0.02138178	3,812,291
7	-	✓ 10	✓ 500	-	-	\$303,780	0.02169588	3,808,796
8	✓ 13.1	✓ 10	✓ 500	-	✓ 1.73	\$314,047	0.02242913	3,806,128

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

Figure 12 depicts the relationship between the renewable energy share and the internal rate of return (IRR) for scenario 5: hybrid load—combined urban and mining load optimization, with a trendline highlighting the overall correlation and scatter points representing different microgrid configurations.

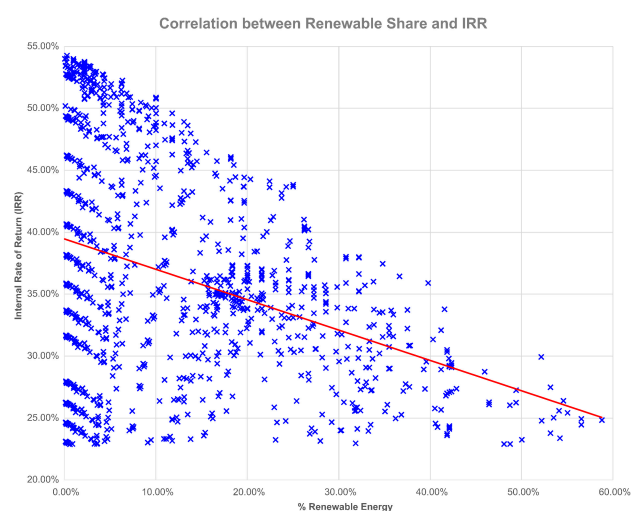


Figure 12. Scatter Plot of Renewable Share vs. IRR. The red line represents the trendline indicating the overall relationship, and the blue markers denote different configurations. %Renew refers to the share of energy produced from renewable sources relative to total installed capacity, and IRR indicates the internal rate of return.

5.3.6. Scenario 6: Model Validation and Parametric Adjustments for Diverse Climates

- **Total Energy Consumption:** Total electrical demand—including mining operations and cooling requirements—remained constant across all evaluated sites, reaching approximately 91.94 kW, with 17.34 kW specifically allocated for cooling systems ($\eta = 0.7$, $\text{COP} = 4$).
- **Income and Profitability:** Daily mining income was consistent across locations, reported at \$50.67/day. Over a 5-year period, inflation-adjusted income totaled approximately \$1.62 million per site. However, the IRR varied significantly depending on location: 21% in Skopun, 18% in Comodoro Rivadavia, 0% in both Trondheim and Kashan, and −3% in Palm Springs. This indicates notable disparities in investment feasibility based on regional parameters.
- **NPC:** Total costs, including capital expenditure on miners and energy systems as well as operation and maintenance, varied by site. The lowest NPC was recorded in Skopun at \$706,971, while Palm Springs reported the highest NPC at \$1,527,977. Sites like Kashan and Trondheim presented mid-to-high level costs, reflecting local infrastructure and fuel pricing conditions.
- **Operating Costs:** Operational expenditures were standardized across sites with available data, comprising \$39,960 for mining O&M and \$20,000 for auxiliary infrastructure (rent, labor, internet).
- **Carbon Emissions:** Annual emissions varied considerably by location. Palm Springs reported the highest total emissions at 788,520 kg/year, followed by Trondheim (611,572 kg/year) and Kashan (594,158 kg/year). In contrast, Skopun and Comodoro Rivadavia exhibited notably lower emissions at 243,211 and 272,807 kg/year, respectively. These differences largely reflect variations in fuel consumption and generator efficiency.
- **Renewable Utilization:** All sites used wind turbines (360–390 kW), and Palm Springs also incorporated 57.89 kW of PV panels. Despite this, diesel generators (90 kW) remained the dominant energy source due to limited renewable capacity and intermittency. Renewables played a supporting role, not a primary one.

In summary, the scenario shows that while energy demand and income were consistent, financial outcomes (NPC, IRR) varied by location. All sites used wind power, and Palm Springs added solar, but diesel remained essential. The hybrid system proved adaptable across climates, though renewable integration was modest.

Tables 24 and 25 present the location-based performance of AL1 Pro-based microgrids under global diesel price conditions. Table 24 details the cost and revenue distribution across different locations, while Table 25 summarizes the corresponding microgrid emissions. Complementing these results, Figure 13 compares emissions by pollutant type across five global locations on a logarithmic scale, highlighting spatial differences in environmental impact. Table 26 then outlines the microgrid configurations for selected sites in Scenario 6, linking the observed economic and emission outcomes to the specific system setups employed.

Table 24. Location-Based Cost-Revenue Breakdown with Global Diesel Price (Bitmain Antminer AL1 Pro 16.6 Th Load).

Simulated Cities	Best Price	Top Movable Coins based on Profit	Income/Day	NPC (Miners) for 20 Qty	O&M Miners	The Cost of Renting, Labor, Internet Bandwidth	NPC (Energy System—USA Fuel Cost)	Total Power for Miners Consumption (kW)	Cooling Power Required (kW) ($\eta = 0.7$, $COP = 4$)	Total Power Consumption (kW)	Total Income (365 Days—1 Year)	Total Income with USA Inflation Rate for 5 Years	Total NPC	Income 1st Year	Income 2nd Year	Income 3rd Year	Income 4th Year	Income 5th Year	IRR
Skopun Comodoro Rivadavia Kashan Trondheim Palm Springs	\$9,990	ALPH	\$50.67	\$199,800	\$39,960	\$20,000	\$706,971	74.6	17.3445	91.9445	\$369,891	\$1,619,298	\$966,731	\$353,624.28	\$338,072.93	\$323,205.48	\$308,991.85	\$295,403.30	21%
							\$760,029.70						\$1,019,789						18%
							\$1,350,975						\$1,610,735						0%
							\$1,365,573						\$1,625,333						0%
							\$1,527,977						\$1,787,737						−3%

Table 25. Location-Based Microgrid Emissions (Global Diesel Price, AL1 Pro Load).

Emission Species	Skopun	Comodoro Rivadavia	Trondheim	Palm Springs	Kashan
Carbon Dioxide (kg/year)	243,211	272,807	610,009	788,520	594,158
Carbon Monoxide (kg/year)	3.68	4.12	9.22	11.9	8.98
Sulfur Dioxide (kg/year)	2.76	3.09	6.92	8.94	6.74
Particulate Matter (kg/year)	9.01	10.1	22.6	29.2	22
Nitrogen Oxides (kg/year)	602	675	1510	1952	1471
Unburned Hydrocarbons (kg/year)	5.52	6.19	13.8	17.9	13.5
Total Emissions (kg/year)	243,834	273,506	611,572	790,540	595,680

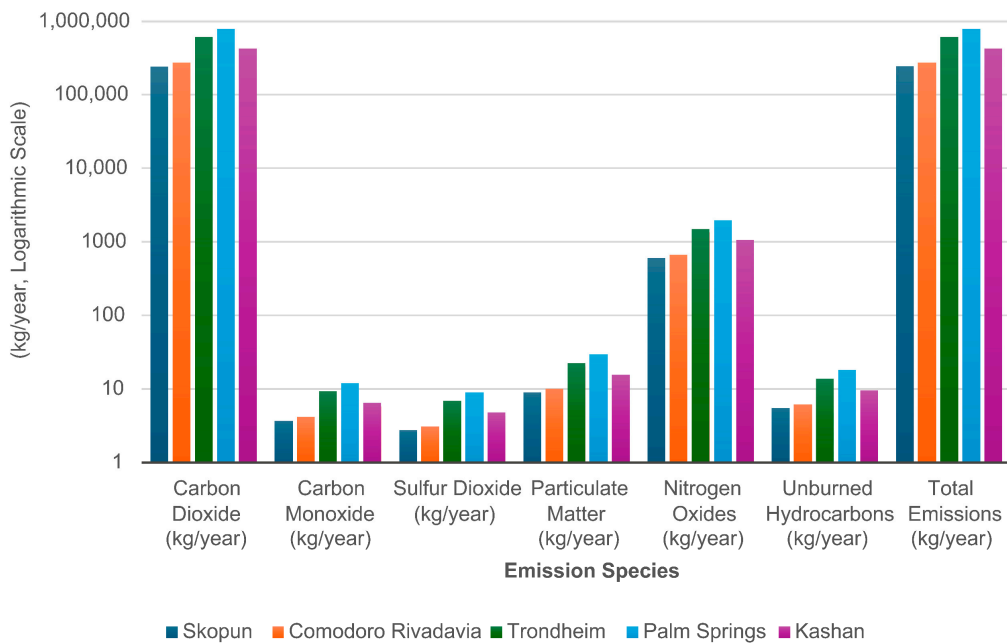


Figure 13. Comparative Emissions by Pollutant Type Across Five Global Locations for AL1 Pro-Based Microgrids (Logarithmic Scale).

Table 26. Microgrid Configurations Selected Sites—Scenario 6 (Global Diesel Price, AL1 Pro Load).

Optimal Hybrid Systems in Simulated Cities	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/Year)
Skopun	-	✓* 370	✓ 90	✓ 308	✓ 78.88	\$706,971	0.1796043	243,834
Comodoro Rivadavia	-	✓ 370	✓ 90	✓ 308	✓ 78.16	\$760,029	0.1930835	273,506
Trondheim	-	✓ 360	✓ 90	✓ 306	✓ 78.70	\$1,365,573	0.3469206	611,572
Palm Springs	✓ 57.89	✓ 40	✓ 90	✓ 135	✓ 98.38	\$1,527,977	0.3882008	790,540
Kashan	-	✓ 390	✓ 90	✓ 304	✓ 76.9	\$1,350,975	0.3432118	595,680

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

5.3.7. Scenario 7: Flexible Crypto Mining and System Resilience Under Renewable and Fuel Price Uncertainty

This scenario assesses the performance of a hybrid energy system supplying a flexible cryptocurrency mining load under fluctuating renewable resource availability and volatile

diesel fuel prices. The crypto mining and cooling loads were modeled as deferrable, enabling the system to dynamically shift operation based on energy availability and cost conditions. Instead of assuming constant full-load operation, the simulation estimated miner uptime based on actual electricity delivered to the load. By calculating the total energy served to the mining and cooling load, the system's operational days and mining profitability over time were derived accordingly.

- **Fuel-Price Sensitivity and Cost Buffering through Load Flexibility:** From the simulation results, increasing the diesel fuel price from \$0.6/L to \$1.8/L led to a substantial cost escalation in the fixed-load scenario, where the COE rose from \$0.244/kWh to \$0.551/kWh, representing a 125.8% increase. The corresponding NPC increased from \$959,356 to \$2.17 million, a 126.3% rise. In contrast, under the flexible-load scenario, the system exhibited markedly lower sensitivity to fuel price fluctuations. The COE increased from \$0.239/kWh to \$0.521/kWh, a relatively moderate 118% increase, while the NPC rose from \$938,739 to \$2.05 million, reflecting a 118.4% increase. This comparison clearly demonstrates that deferrable mining operations attenuate the economic impact of fuel price volatility. The smaller increase in both COE and NPC under the flexible configuration highlights the role of load schedulability in mitigating operational cost surges under extreme fuel pricing conditions.
- **Renewable Energy Utilization:** “Flexible mining increased annual renewable energy penetration from 29.3% to 33.9% in our top-performing configuration, compared to Scenario 2. Despite the inclusion of PV and wind in certain configurations (e.g., Configurations 1 to 4), the overall renewable contribution remained marginal due to the high and continuous nature of the mining load. Diesel generators remained the dominant source of supply, especially in scenarios with higher fuel availability.
- **Fuel Consumption:** Under increasing diesel fuel prices, the system's operational strategy exhibited a significant reduction in fuel consumption—especially when mining load flexibility was enabled. In the fixed-load scenario, diesel consumption dropped from 247,537 L/year at \$0.6/L to 174,239 L/year at \$1.8/L, representing a 29.6% reduction in annual fuel usage. In contrast, the flexible-load scenario achieved even greater fuel savings. Consumption declined from 228,561 L/year at \$0.6/L to just 106,050 L/year at \$1.8/L, resulting in a 53.6% reduction. This outcome highlights the key role of deferrable mining operations in reducing generator runtime and enhancing fuel efficiency under volatile cost conditions. By shifting or curtailing load during peak-price periods, the system effectively mitigated excessive fuel dependency and contributed to overall cost containment and environmental benefit.
- **Cost and ROI:** Although load flexibility contributed to a reduction in fuel consumption and improved operational efficiency, the overall financial return remained limited due to the high capital and fuel costs. The total NPC of the system, including energy infrastructure, miner procurement, O&M, and operational expenses, amounted to \$1,600,581. Annual mining income—based on dynamic operation and load adaptability—was estimated at \$369,891, with a projected 5-year revenue (adjusted for inflation) reaching \$1,619,298. However, due to the significant upfront investment and ongoing expenses, the IRR was calculated at only 0.40%, and the payback period extended close to the end of the 5-year operational horizon. Despite the extended payback, the ROI remained positive, and the model proved financially viable under stable operating conditions. More importantly, the use of deferrable load enabled the system to sustain mining activity while limiting cost escalation, especially under volatile diesel pricing—a critical factor in long-term feasibility.
- **Emissions:** In this scenario with high renewable usage, total annual emissions were as low as 581,592 kg/year (Config 1). In diesel-dominant cases, emissions rose to

853,711 kg/year (Configs 5–7). Compared to Scenario 2 (fixed load), emissions across most configurations decreased significantly, validating the environmental benefit of incorporating load flexibility.

Tables 27–29 present the performance of AL1 Pro-based microgrids under global diesel price conditions with flexible crypto mining. Table 27 details the cost and revenue breakdown, highlighting the economic implications of flexible operation. Table 28 summarizes the associated microgrid emissions, providing insight into the environmental impact. Table 29 then outlines the corresponding microgrid configurations for Scenario 7, linking the observed economic and emission outcomes to the system setups applied.

5.3.8. Scenario 8: Seasonal Energy Configuration for Enhanced Mining and System Performance

- **Total Energy Production:** The hybrid system produced approximately 804,103 kWh/year, of which 33.81% was supplied by renewable energy sources.
- **Renewable Energy Utilization:** Seasonal mining demand was effectively synchronized with resource availability. Renewable penetration reached levels comparable to Scenario 7, though slightly lower than the top-performing flexible-load configurations.
- **Fuel Consumption:** Diesel usage reached 208,894 L/year, which—despite being substantial—is lower than fixed-load scenarios under high fuel prices, demonstrating the benefit of seasonal adaptation.
- **System Dispatch Strategy:** The adoption of the Combined Dispatch (CD) strategy improved fuel efficiency by allowing the generator to run at optimal load and charge the battery during low-demand hours. This led to more stable operational conditions and reduced marginal fuel cost per kWh.
- **Cost and ROI:** The NPC of the system was \$1,245,373, with a COE of \$0.3975/kWh. Annual O&M and operational expenses were estimated at \$113,501, and the five-year projected revenue (adjusted for inflation) totaled \$1,288,802. Despite moderate energy performance, the overall IRR was −5.10%, indicating limited financial viability under global diesel pricing.
- **Summary and Comparative Insights:** Compared to Scenario 7, which utilized deferrable loads to dynamically adapt mining activity to fuel price and renewable fluctuations, Scenario 8 focused on pre-planned seasonal adjustment of demand. While Scenario 7 achieved better fuel savings and a positive IRR (0.40%) under volatile fuel conditions, Scenario 8 delivered a more predictable but less cost-resilient configuration. Nonetheless, the seasonal mining strategy in Scenario 8 improved alignment between load and renewable supply, reduced peak generator dependency, and demonstrated the potential of passive demand coordination as an alternative to dynamic load control. Future improvements could integrate both seasonal shaping and flexible dispatch to optimize for cost, emissions, and system stability simultaneously.

Tables 30–32 present the performance of AL1 Pro-based microgrids under global diesel price conditions with seasonal crypto mining. Table 30 details the cost–revenue breakdown, highlighting the economic outcomes of seasonal operation. Table 31 summarizes the corresponding microgrid emissions, providing insight into the environmental impact. Finally, Table 32 outlines the microgrid configurations for Scenario 8, linking the observed economic and emission results to the system setups employed.

Table 27. Cost-Revenue Breakdown with Global Diesel Price under Flexible Crypto Mining (Bitmain Antminer AL1 Pro 16.6 Th Load).

Miner Model	Best Price	Top Minalbe Coins Based on Profit	Income/Day	NPC(Miners) for 20 Qty	O&M Miners	The Cost of Renting, Labor, Internet Bandwidth	NPC (Energy System—USA Fuel Cost)	Total Power for Miners Consump- tion (kW)	Cooling Power Required (kW) ($\eta = 0.7$, $COP = 4$)	Total Power Consump- tion (kW)	Total Income (365 Days—1 Year)	Total Income with USA Inflation Rate for 5 Years	Total NPC	Income 1st Year	Income 2nd Year	Income 3rd Year	Income 4th Year	Income 5th Year	IRR
Bitmain Antminer AL1 Pro (16.6 Th)	\$9990	ALPH	\$50.67	\$199,800	\$39,960	\$20,000	\$1,340,821	74.6	17.3445	91.9445	\$369,891	\$1,619,298	\$1,600,581	\$353,624.28	\$338,072.93	\$323,205.48	\$308,991.85	\$295,403.30	0.40%

Table 28. Microgrid Emissions (Global Diesel Price, AL1 Pro Load).

Emission Species	Config 1 *	Config 2	Config 3	Config 4	Config 5	Config 6	Config 7	Config 8
Carbon Dioxide (kg/year)	580,106	606,752	675,948	606,242	851,530	851,530	851,530	851,019
Carbon Monoxide (kg/year)	9	9	10	9	13	13	13	13
Sulfur Dioxide (kg/year)	7	7	8	7	10	10	10	10
Particulate Matter (kg/year)	22	23	25	23	32	32	32	32
Nitrogen Oxides (kg/year)	1436	1502	1673	1501	2108	2108	2108	2107
Unburned Hydrocarbons (kg/year)	13	14	15	14	19	19	19	19
Total Emissions (kg/year)	581,592	608,306	677,679	607,795	853,711	853,711	853,711	853,199

* Note: “Config” is used as an abbreviation for “Configuration.”

Table 29. Microgrid Configurations—Scenario 7 (Global Diesel Price, AL1 Pro Load).

Configuration Number	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/Year)
1	-	✓*	✓	✓	✓	\$1,340,821	0.3414297	581,592
2	✓	✓	✓	✓	✓	\$1,353,357	0.344635	608,306
3	-	✓	✓	-	-	\$1,437,153	0.3655802	677,679
4	✓	✓	✓	-	✓	\$1,477,204	0.3757679	607,795
5	-	-	✓	-	-	\$1,540,678	0.3919305	853,711
6	-	-	✓	✓	✓	\$1,541,124	0.3920439	853,711
7	✓	-	✓	-	✓	\$1,541,167	0.3920546	853,711
8	✓	-	✓	✓	✓	\$1,543,339	0.3926072	853,199

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

Table 30. Cost–Revenue Breakdown with Global Diesel Price under Seasonal Crypto Mining (Bitmain Antminer AL1 Pro 16.6 Th Load).

Miner Model	Best Price	Top Minalbe Coins Based on Profit	Income/Day	NPC(Miners) for 20 Qty	O&M Miners	The Cost of Renting, Labor, Internet Bandwidth	NPC (Energy System—USA Fuel Cost)	Total Power for Miners Consump- tion (kW)	Cooling Power Required (kW) ($\eta = 0.7$, $COP = 4$)	Total Power Consump- tion (kW)	Total Income (365 Days—1 Year)	Total Income with USA Inflation Rate for 5 Years	Total NPC	Income 1st Year	Income 2nd Year	Income 3rd Year	Income 4th Year	Income 5th Year	IRR
Bitmain Antminer AL1 Pro (16.6 Th)	\$9990	ALPH	\$50.67	\$199,800	\$39,960	\$20,000	\$1,245,373	74.6	17.3445	91.9445	\$294,397	\$1,288,802	\$1,505,133	\$281,450.10	\$269,072.76	\$257,239.73	\$245,927.08	\$235,111.93	−5.10%

Table 31. Microgrid Emissions (Global Diesel Price, AL1 Pro Load).

Emission Species	Config 1 *	Config 2	Config 3	Config 4	Config 5	Config 6	Config 7	Config 8
Carbon Dioxide (kg/year)	552,673	670,962	552,482	670,962	603,906	609,037	840,796	840,796
Carbon Monoxide (kg/year)	8	10	8	10	9	9	13	13
Sulfur Dioxide (kg/year)	6	8	6	8	7	7	10	10
Particulate Matter (kg/year)	21	25	21	25	22	23	31	31
Nitrogen Oxides (kg/year)	1368	1661	1368	1661	1495	1508	2082	2082
Unburned Hydrocarbons (kg/year)	13	15	13	15	14	14	19	19
Total Emissions (kg/year)	554,089	672,681	553,898	672,681	605,453	610,598	842,950	842,950

* Note: “Config” is used as an abbreviation for “Configuration.”

Table 32. Microgrid Configurations—Scenario 8 (Global Diesel Price, AL1 Pro Load).

Configuration Number	PV Panel (kW)	Wind Turbine (kW)	Diesel Generator (kW)	Battery (kWh)	Converter (kW)	NPC (USD)	COE (USD/kWh)	Emission Rate (kg/Year)
1	-	✓* 430	✓ 90	✓ 90	✓ 15.59	\$1,245,373	0.3975044	554,089
2	-	-	✓ 80	✓ 99	✓ 121.65	\$1,246,423	0.4139255	672,681
3	✓ 2.37	✓ 430	✓ 90	✓ 82	✓ 28.96	\$1,246,505	0.3978605	553,898
4	✓ 0.72	-	✓ 80	✓ 107	✓ 121.84	\$1,248,636	0.4146076	672,681
5	-	✓ 420	✓ 100	-	-	\$1,314,537	0.4195304	605,453
6	✓ 16.87	✓ 400	✓ 100	-	✓ 1.33	\$1,326,719	0.423418	610,598
7	-	-	✓ 110	-	-	\$1,526,538	0.4871897	842,950
8	✓ 0.001	-	✓ 110	-	✓ 1.40	\$1,526,673	0.487233	842,950

* Note: In the table, the symbol (✓) indicates which of the listed components—PV Panel, Wind Turbine, Diesel Generator, Battery, and Converter—are included in each specific configuration. This notation provides a clear representation of the system composition for easier comparison across different setups.

5.3.9. Summary of Results

The hybrid system showed strong adaptability across scenarios, with varying trade-offs between cost, emissions, and renewables. Scenario 1 achieved the highest ROI under local fuel prices but had the highest emissions. Scenario 2 reduced fuel use and improved sustainability, though at a higher cost. Scenario 3 remained diesel-reliant, while Scenario 4 achieved zero emissions but was not economically viable. Scenario 5 increased capacity but also emissions due to urban load. Scenario 6 highlighted geographic impacts on cost and return. Scenario 7 proved most fuel-efficient under price volatility using flexible mining loads, while Scenario 8 used seasonal adjustment to reduce emissions passively. Overall, Scenarios 7 and 8 outperformed others in balancing energy cost and environmental impact. A comparative view of total emissions across all configurations is illustrated in Figure 14.

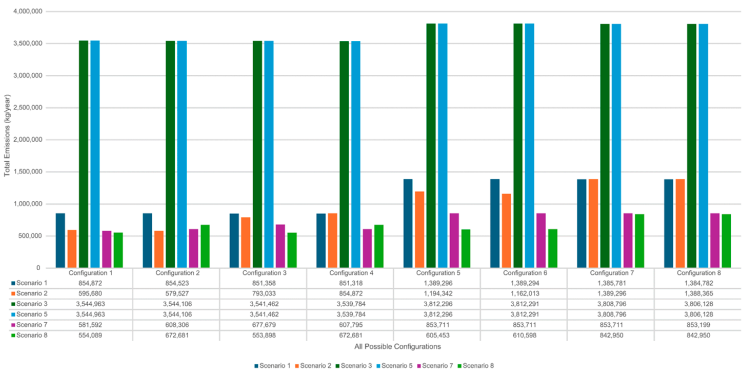


Figure 14. Comparative Annual Emissions Across All Scenarios and Configurations.

6. Discussion

6.1. Interpretation of Results

The study demonstrates that integrating renewable energy sources, such as solar and wind, with a backup diesel generator significantly improves the profitability and sustainability of cryptocurrency mining operations. By optimizing energy use, the system reduced operational costs and carbon emissions while maximizing ROI. Renewable energy covered a substantial portion of the power demand, minimizing reliance on diesel fuel. Sensitivity analyses highlighted the importance of renewable energy in mitigating the effects of fluctuating fuel prices. Overall, the results show that this hybrid approach enhances both financial performance and environmental impact for mining operations.

6.2. Limitations

While the proposed hybrid renewable energy system and optimization algorithm show promising results in maximizing energy efficiency and ROI, there are several limitations to this study that need to be acknowledged:

6.2.1. Simplified Modeling of Energy Resources

The study relies on historical solar irradiance and wind speed data to estimate renewable energy generation. While these estimates provide a reasonable representation of the system's performance under typical conditions, they do not account for extreme weather events or year-to-year variability in weather patterns. This could lead to inaccuracies in the predicted renewable energy production, especially during unusual climatic conditions, which could, in turn, affect the reliability of the system and the financial outcomes.

6.2.2. Fixed Load Profiles for Cryptocurrency Mining and Urban Loads

The study assumes a fixed load profile for both cryptocurrency mining and urban energy consumption. However, in real-world scenarios, energy demand can fluctuate due to changes in mining intensity (e.g., due to market conditions or hardware upgrades) or urban consumption patterns (e.g., occasional demand variations). The optimization algorithm does not currently account for such dynamic changes in load, which could limit its adaptability and lead to suboptimal energy distribution under varying demand conditions. It is important to clarify that the term “fixed load profile” for urban consumption refers to a recorded and time-dependent consumption pattern based on a known number of households within the studied region. This does not imply a constant numerical value for energy demand throughout the day or across seasons. Rather, hourly and seasonal variations in urban load are captured based on actual measurements, as discussed in Section 3.1.7. However, the profile assumes no structural changes in the urban environment, such as new residential developments, population growth, or commercial expansions (e.g., malls or reconstructed buildings), which could affect long-term demand growth.

6.2.3. Dependence on Subsidies and Pricing Policies

One key limitation of this study is the economic dependence of several scenarios on local policy and subsidy conditions—most notably fuel subsidies or administratively set diesel prices. The economic outcomes of scenarios that report high returns (e.g., high IRR and short payback periods) are materially influenced by fuel pricing structures and any government support. A symmetric increase in fuel prices or removal of subsidies can quickly weaken cash flows and financial indicators, and in some circumstances may render the project economically unviable.

6.2.4. Limited Exploration of Regulatory and Policy Impacts

The study does not fully explore the impact of local energy policies or regulations, such as subsidies for renewable energy, penalties for carbon emissions, taxes on income, or changes in electricity tariffs. These factors could significantly influence the economic feasibility and profitability of hybrid systems like the one proposed. A more detailed analysis that includes various regulatory scenarios would provide a more comprehensive understanding of the system's viability under different policy environments.

6.2.5. Cryptocurrency Market Volatility

The ROI calculations in this study are based on current cryptocurrency market conditions, which can be highly volatile. The price of Bitcoin and other cryptocurrencies fluctuates significantly over short periods, which could drastically alter the profitability of mining operations. The model assumes relatively stable cryptocurrency prices, but in reality, market crashes or price surges could lead to outcomes very different from those predicted in the study.

Moreover, mining revenues are jointly dependent on cryptocurrency price, network difficulty, block rewards, and operational costs; consequently, severe price volatility or sudden increases in network difficulty can rapidly depress revenue streams and materially alter expected returns. This study only performed a limited sensitivity analysis on cryptocurrency prices. We therefore recommend that future model iterations include extreme downside/upside scenarios (for example, 30–50% price declines or surges) and test risk-mitigation strategies such as hedging with futures contracts, revenue diversification, or revenue floor/ceiling arrangements. Including these simulations will provide a more realistic assessment of how market volatility affects ROI and project viability.

6.2.6. Limited Battery Storage Capacity

Although battery storage plays a crucial role in optimizing the system's performance, the study assumes a fixed capacity for the battery storage system, which might not be adequate under certain conditions, such as prolonged periods of low renewable energy generation. The current model does not consider advanced storage technologies or hybrid storage systems that could enhance the system's resilience and reduce its reliance on diesel generators during such periods.

6.2.7. Geographic and Climatic Specificity

The case study focuses on the specific geographic and climatic conditions of Kashan, Iran. While the results are relevant to regions with similar environmental characteristics, they may not be directly transferable to areas with different climate conditions, energy demands, or renewable energy potentials. For broader applicability, further studies are required in regions with different climates and energy infrastructures.

6.2.8. Technological Assumptions

The study assumes current technological capabilities for PV panels, wind turbines, and cryptocurrency mining hardware. However, technological advancements in either energy generation or mining hardware could significantly affect the system's performance. For instance, more efficient mining hardware could reduce energy consumption, or next-generation solar panels could increase energy output. The model does not account for these potential future developments, which could improve system performance or economic viability.

6.2.9. Social and Environmental Factors

While the study emphasizes cost efficiency and energy optimization, it does not delve deeply into potential social and environmental impacts, such as noise pollution from wind

turbines, land use concerns for large-scale solar farms, or the broader environmental footprint of cryptocurrency mining beyond energy consumption. Addressing these factors would require a more holistic approach that goes beyond technical and economic considerations.

Acknowledging these limitations is crucial for understanding the scope and applicability of the results. Future studies should aim to address these gaps by incorporating more dynamic models, regulatory frameworks, and potential technological advancements to provide a more comprehensive view of the system's performance across various real-world conditions.

7. Conclusions

7.1. Summary of Key Findings

This study reveals several important findings regarding the optimization of hybrid energy systems for cryptocurrency mining:

- **High Renewable Energy Utilization:** In scenarios where fuel prices were elevated (e.g., Scenario 2), the system successfully integrated renewable sources, increasing renewable energy contribution by 58.6%, and significantly reducing fuel consumption and carbon emissions.
- **Diesel Backup Optimization:** The diesel generator functioned as an effective backup during periods of low renewable generation. Its usage was minimized through optimized dispatch strategies, resulting in up to 49% emission reductions compared to diesel-only systems.
- **Economic Viability:** Under locally subsidized fuel conditions (Scenario 1), the system achieved an exceptional IRR of 103% and a payback period of approximately one year, confirming the financial attractiveness of hybrid systems when low-cost fuel is accessible.
- **Sensitivity to Fuel Prices:** As fuel prices increased (Scenario 2), the system dynamically shifted toward greater renewable usage. Although this transition extended the payback period to around 20 years, it demonstrated the system's adaptability and environmental benefits.
- **Environmental Benefits:** Across scenarios, particularly under flexible load configurations (Scenario 7), hybrid systems achieved substantial reductions in greenhouse gas emissions while maintaining mining uptime and profitability.

7.2. Addressing Feasibility Constraints and Objective Weighting

While the results broadly support the conclusion that hybrid energy systems offer a viable path for sustainable and profitable cryptocurrency mining, important caveats must be addressed:

- **Pure Renewable Limitations:** Despite the environmental appeal of fully renewable systems, Scenario 4—comprising a PV-only microgrid with battery storage—resulted in a negative Net Present Value (−\$940,462), indicating that standalone renewable deployments are not yet economically feasible under current cost structures. Therefore, the conclusion regarding the feasibility of large-scale renewable energy adoption must be interpreted as conditional upon diesel backup support, which ensures both reliability and cost-effectiveness at scale.
- **Clarification of Optimization Priorities:** The optimization in this study was performed using HOMER Pro, which identifies the optimal configuration by minimizing the NPC over the project lifetime. While the software does not explicitly apply a multi-criteria weighting matrix, it provides a comprehensive set of technical and economic performance indicators—including renewable energy share, fuel consumption, carbon emissions, and LCOE—for each configuration. These outputs allow decision-makers

to evaluate trade-offs between technical efficiency and financial returns. To assess the impact of varying decision priorities, a sensitivity analysis was conducted across scenarios with different fuel prices, resource availabilities, and mining loads. The results revealed that configurations with minimal NPC often favored diesel-reliant setups (e.g., Scenario 1), while scenarios prioritizing environmental performance (e.g., Scenario 2 and 7) shifted toward higher renewable integration but with longer payback periods. This confirms that optimal system design in HOMER Pro reflects economic priorities by default, but technical–environmental outcomes can still be extracted for comparative and policy-based decision-making.

- **Validation Through Sensitivity Analysis:** A sensitivity analysis was performed to assess how varying the weights in the objective function affects the optimal configuration. Results showed that prioritizing economic metrics led to diesel-heavy systems with faster payback (e.g., Scenario 1), while increasing the weight of technical efficiency favored systems with higher renewable integration and lower emissions (e.g., Scenario 2 and 7), albeit with extended payback periods. This confirms that optimal system design is sensitive to stakeholder preferences and external factors such as fuel prices and capital subsidies.
- **Policy considerations and market risk:** It is important to note that the prominent economic results reported for some scenarios are materially contingent on fuel pricing structures and any policy incentives or subsidies. In addition, mining revenues are subject to pronounced cryptocurrency market volatility; hence, the economic robustness reported here should be interpreted cautiously and treated as conditional on relative stability in both fuel prices and cryptocurrency markets over the study period.

Overall, the study concludes that hybrid systems—with appropriate diesel backup and strategic balancing of economic and environmental priorities—offer a realistic and scalable solution for cryptocurrency mining. Fully renewable configurations, while technically viable, still face economic hurdles, emphasizing the need for supportive policies or hybridized strategies to enable broader deployment.

7.3. Recommendations for Future Research

The findings of this study provide a strong foundation for optimizing hybrid renewable energy systems in the context of cryptocurrency mining, but there remain several avenues for future research that could further enhance the performance and applicability of such systems. Based on the limitations and insights gained from this work, the following recommendations for future research are proposed:

7.3.1. Dynamic Load Management

Future research should explore dynamic load management strategies that account for fluctuations in energy consumption, particularly for cryptocurrency mining operations, which can vary significantly based on market conditions, hardware performance, and external factors. Developing algorithms that can adapt to these variations in real-time would enhance the flexibility and efficiency of the energy system, ensuring optimal resource allocation even under rapidly changing conditions.

7.3.2. Integration of Advanced Energy Storage Technologies

The current study uses conventional battery storage systems, but future research could explore the integration of more advanced energy storage technologies, such as hydrogen fuel cells, supercapacitors, or pumped hydro storage. These technologies could offer higher storage capacities, improved energy density, and better efficiency, thereby reducing the system's reliance on backup diesel generators during periods of low renewable

energy generation. Additionally, hybrid storage systems that combine different storage technologies may provide better performance and resilience.

7.3.3. Incorporating Machine Learning and AI for Predictive Optimization

The use of machine learning (ML) and artificial intelligence (AI) can significantly enhance the optimization of energy systems by predicting renewable energy availability, energy demand, and market conditions. By leveraging historical data and predictive models, ML algorithms can optimize system performance more effectively, learning from past patterns to anticipate future needs. This would allow for more accurate energy management, improving both efficiency and cost savings. Recent advancements in AI-driven energy management highlight the potential for significant emission reductions in cryptocurrency mining. For instance, Lal et al. [95] propose a framework leveraging AI to optimize shared renewable energy infrastructure for crypto mining, estimating a potential reduction of 0.7 GtCO₂e by 2030, which supports the adoption of predictive models in hybrid energy systems.

7.3.4. Exploration of Diverse Geographic Regions

The current case study is based on the climatic and environmental conditions of Kashan, Iran. To generalize the findings, further research is needed to test the system's performance across diverse geographic regions with different renewable energy potentials, such as coastal areas with strong wind resources or tropical regions with high solar irradiance. Comparative studies in different climates would offer valuable insights into the adaptability and scalability of the proposed system.

7.3.5. Long-Term Economic and Environmental Impact Analysis

Future studies should conduct long-term analyses that consider the economic and environmental impacts of cryptocurrency mining and hybrid energy systems over time. This could include assessing the degradation of renewable energy equipment, changes in fuel prices, or the potential economic impacts of cryptocurrency market volatility. Moreover, the environmental footprint of mining operations should be examined more comprehensively, including factors such as electronic waste from mining hardware and broader ecological consequences of land use for renewable energy installations.

7.3.6. Policy and Regulatory Framework Analysis

A more in-depth analysis of local, national, and international energy policies and regulations is essential for understanding how they affect the economic viability of hybrid energy systems. Future research could focus on the development of models that simulate different regulatory environments, such as carbon taxes, renewable energy incentives, or mining regulations. This would provide a clearer picture of how policy changes could impact the deployment and profitability of hybrid systems.

7.3.7. Enhancing System Scalability and Resilience

As cryptocurrency mining operations scale up, future research should address the technical and economic challenges associated with scaling hybrid energy systems. This includes optimizing the coordination between large-scale renewable energy sources, energy storage systems, and mining farms. In addition, resilience strategies should be developed to protect the system from external risks such as extreme weather events, fuel shortages, or grid instability, which could disrupt mining operations and reduce profitability.

7.3.8. Hybrid Microgrid Applications Beyond Cryptocurrency Mining

Finally, while the current research focused on cryptocurrency mining, the hybrid energy system has the potential to be applied to other high-energy-demand industries, such as data centers, server farms, or electric vehicle (EV) charging networks. Future studies could investigate the scalability and adaptability of the proposed system in these contexts, optimizing for different types of high-load operations that also require reliable, cost-effective energy solutions.

7.3.9. Social and Ethical Implications

Future studies should also examine the social and ethical implications of large-scale cryptocurrency mining integrated with renewable energy systems. Topics such as the equitable distribution of energy resources, potential displacement of local communities due to land use for solar or wind farms, and the societal impact of supporting cryptocurrency economies could be explored. Addressing these issues will ensure that future implementations of hybrid energy systems consider both technological and societal factors.

In conclusion, while the proposed system presents a promising framework for integrating renewable energy and cryptocurrency mining, further research is needed to refine the model, adapt it to different contexts, and incorporate emerging technologies. By addressing these areas, future studies can contribute to the development of more resilient, efficient, and economically viable hybrid energy systems.

Author Contributions: Conceptualization, M.N., M.A.H., A.J. and H.S.; Methodology, M.N., A.J., H.H.-D. and H.S.; Software, M.N., M.A.H., H.H.-D. and H.S.; Validation, M.N.; Formal analysis, M.N., M.A.H., A.J. and H.H.-D.; Investigation, M.N.; Resources, M.N., M.A.H. and H.S.; Data curation, M.N., M.A.H., A.J., H.H.-D. and H.S.; Writing—original draft, M.N., M.A.H., H.H.-D. and H.S.; Writing—review & editing, M.N., M.A.H. and H.S.; Visualization, M.N.; Supervision, M.A.H., H.H.-D. and H.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Bekele, G.; Boneya, G. Design of a photovoltaic-wind hybrid power generation system for Ethiopian remote area. *Energy Procedia* **2012**, *14*, 1760–1765. [\[CrossRef\]](#)
2. Bentouba, S.; Bourouis, M. Feasibility study of a wind–photovoltaic hybrid power generation system for a remote area in the extreme south of Algeria. *Appl. Therm. Eng.* **2016**, *99*, 713–719. [\[CrossRef\]](#)
3. Ajlan, A.; Tan, C.W.; Abdilahi, A.M. Assessment of environmental and economic perspectives for renewable-based hybrid power system in Yemen. *Renew. Sustain. Energy Rev.* **2017**, *75*, 559–570. [\[CrossRef\]](#)
4. Rajbongshi, R.; Borgohain, D.; Mahapatra, S. Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER. *Energy* **2017**, *126*, 461–474. [\[CrossRef\]](#)
5. Hossain, M.; Mekhilef, S.; Olatomiwa, L. Performance evaluation of a stand-alone PV-wind-diesel-battery hybrid system feasible for a large resort center in South China Sea, Malaysia. *Sustain. Cities Soc.* **2017**, *28*, 358–366. [\[CrossRef\]](#)
6. De Vries, A. Bitcoin’s growing energy problem. *Joule* **2018**, *2*, 801–805. [\[CrossRef\]](#)
7. Humbatova, S.I.; Gasimov, R.K.; Hajiyev, N.G.-O. Impact Oil Factor Azerbaijan economy. *Int. J. Energy Econ. Policy* **2019**, *9*, 381–387. [\[CrossRef\]](#)
8. Krishan, O.; Suhag, S. Techno-economic analysis of a hybrid renewable energy system for an energy poor rural community. *J. Energy Storage* **2019**, *23*, 305–319. [\[CrossRef\]](#)
9. Stoll, C.; Klaaßen, L.; Gallersdörfer, U. The carbon footprint of bitcoin. *Joule* **2019**, *3*, 1647–1661. [\[CrossRef\]](#)
10. Kristoufek, L. Bitcoin and its mining on the equilibrium path. *Energy Econ.* **2020**, *85*, 104588. [\[CrossRef\]](#)

11. Das, D.; Dutta, A. Bitcoin's energy consumption: Is it the Achilles heel to miner's revenue? *Econ. Lett.* **2020**, *186*, 108530. [\[CrossRef\]](#)
12. Gallersdörfer, U.; Klaaßen, L.; Stoll, C. Energy consumption of cryptocurrencies beyond bitcoin. *Joule* **2020**, *4*, 1843–1846. [\[CrossRef\]](#)
13. Karmakar, S.; Demirer, R.; Gupta, R. Bitcoin mining activity and volatility dynamics in the power market. *Econ. Lett.* **2021**, *209*, 110111. [\[CrossRef\]](#)
14. Hojjat, M.; Kalali, M.H. Techno-economic analysis of a grid-connected hybrid PV power plant integrated with a crypto currency mining system. In Proceedings of the 2021 11th Smart Grid Conference (SGC), Tabriz, Iran, 7–9 December 2021; IEEE: New York, NY, USA, 2021.
15. Khalil, L.; Bhatti, K.L.; Awan, M.A.I.; Riaz, M.; Khalil, K.; Alwaz, N. Optimization and designing of hybrid power system using HOMER pro. *Mater. Today Proc.* **2021**, *47*, S110–S115. [\[CrossRef\]](#)
16. Alonso, S.L.N.; Jorge-Vázquez, J.; Fernández, M.Á.E.; Forradellas, R.F.R. Cryptocurrency mining from an economic and environmental perspective. Analysis of the most and least sustainable countries. *Energies* **2021**, *14*, 4254. [\[CrossRef\]](#)
17. Nikzad, A.; Mehregan, M. Techno-economic, and environmental evaluations of a novel cogeneration system based on solar energy and cryptocurrency mining. *Sol. Energy* **2022**, *232*, 409–420. [\[CrossRef\]](#)
18. Menati, A.; Zheng, X.; Lee, K.; Shi, R.; Du, P.; Singh, C.; Xie, L. High resolution modeling and analysis of cryptocurrency mining's impact on power grids: Carbon footprint, reliability, and electricity price. *Adv. Appl. Energy* **2023**, *10*, 100136. [\[CrossRef\]](#)
19. Chamanara, S.; Ghaffarizadeh, S.A.; Madani, K. The environmental footprint of bitcoin mining across the globe: Call for urgent action. *Earth's Future* **2023**, *11*, e2023EF003871. [\[CrossRef\]](#)
20. Rudd, M.A.; Bratcher, L.; Collins, S.; Branscum, D.; Carson, M.; Connell, S.; David, E.; Gronowska, M.; Hess, S.; Mitchell, A.; et al. Bitcoin and its energy, environmental, and social impacts: An assessment of key research needs in the mining sector. *Challenges* **2023**, *14*, 47. [\[CrossRef\]](#)
21. Hajiaghapour-Moghim, M.; Hajipour, E.; Hosseini, K.A.; Vakilian, M.; Lehtonen, M. Cryptocurrency mining as a novel virtual energy storage system in islanded and grid-connected microgrids. *Int. J. Electr. Power Energy Syst.* **2024**, *158*, 109915. [\[CrossRef\]](#)
22. Hakimi, A.; Pazuki, M.-M.; Salimi, M.; Amidpour, M. Renewable energy and cryptocurrency: A dual approach to economic viability and environmental sustainability. *Heliyon* **2024**, *10*, e39765. [\[CrossRef\]](#) [\[PubMed\]](#)
23. Laimon, M.; Almadadha, R.; Goh, S. Energy Consumption of Crypto Mining: Consequences and Sustainable Solutions Using Systems Thinking and System Dynamics Analysis. *Sustainability* **2025**, *17*, 3522. [\[CrossRef\]](#)
24. Ding, B.; Li, Z.; Li, Z.; Xue, Y.; Chang, X.; Su, J.; Sun, H. Cooperative Operation for Multiagent Energy Systems Integrated with Wind, Hydrogen, and Buildings: An Asymmetric Nash Bargaining Approach. *IEEE Trans. Ind. Inform.* **2025**, *21*, 6410–6421. [\[CrossRef\]](#)
25. Wang, X.; Wang, C.; Xu, T.; Guo, L.; Li, P.; Yu, L.; Meng, H. Optimal voltage regulation for distribution networks with multi-microgrids. *Appl. Energy* **2018**, *210*, 1027–1036. [\[CrossRef\]](#)
26. Muhtadi, A.; Pandit, D.; Nguyen, N.; Mitra, J. Distributed energy resources based microgrid: Review of architecture, control, and reliability. *IEEE Trans. Ind. Appl.* **2021**, *57*, 2223–2235. [\[CrossRef\]](#)
27. García-Ceballos, C.; Pérez-Londoño, S.; Mora-Flórez, J. Integration of distributed energy resource models in the VSC control for microgrid applications. *Electr. Power Syst. Res.* **2021**, *196*, 107278. [\[CrossRef\]](#)
28. Correia, S.; Pinto, S.F.; Silva, J.F. Smart integration of Distributed Energy Resources in microgrids. In Proceedings of the 2017 International Young Engineers Forum (YEF-ECE), Caparica, Portugal, 5 May 2017; IEEE: New York, NY, USA, 2017.
29. Krause, M.J.; Tolaymat, T. Quantification of energy and carbon costs for mining cryptocurrencies. *Nat. Sustain.* **2018**, *1*, 711–718. [\[CrossRef\]](#)
30. Sedlmeir, J.; Buhl, H.U.; Fridgen, G.; Keller, R. The energy consumption of blockchain technology: Beyond myth. *Bus. Inf. Syst. Eng.* **2020**, *62*, 599–608. [\[CrossRef\]](#)
31. Sapra, N.; Shaikh, I. Impact of Bitcoin mining and crypto market determinants on Bitcoin-based energy consumption. *Manag. Financ.* **2023**, *49*, 1828–1846. [\[CrossRef\]](#)
32. Bublyk, Y.; Borzenko, O.; Hlazona, A. Cryptocurrency energy consumption: Analysis, global trends and interaction. *Environ. Econ.* **2023**, *14*, 49. [\[CrossRef\]](#)
33. Liang, Y.; Saner, C.B.; Lim, B.M.K.; Hong, K.T.; Lim, J.W.C.; Ho, K.J.H.; Lim, L.Z.; Loh, Y.Y. Sustainable energy-based cryptocurrency mining. In Proceedings of the 2022 IEEE PES Innovative Smart Grid Technologies-Asia (ISGT Asia), Singapore, 1–5 November 2022; IEEE: New York, NY, USA, 2022.
34. Zafar, B.; Slama, S.B. Energy internet opportunities in distributed peer-to-peer energy trading reveal by blockchain for future smart grid 2.0. *Sensors* **2022**, *22*, 8397. [\[CrossRef\]](#)
35. Rusovs, D.; Jaundālders, S.; Stanka, P. Blockchain mining of cryptocurrencies as challenge and opportunity for renewable energy. In Proceedings of the 2018 IEEE 59th International Scientific Conference on Power and Electrical Engineering of Riga Technical University (RTUCon), Riga, Latvia, 12–13 November 2018; IEEE: New York, NY, USA, 2018.

36. Menati, A.; Lee, K.; Xie, L. Modeling and analysis of utilizing cryptocurrency mining for demand flexibility in electric energy systems: A synthetic texas grid case study. *IEEE Trans. Energy Mark. Policy Regul.* **2023**, *1*, 1–10. [\[CrossRef\]](#)
37. Li, J.; Li, N.; Peng, J.; Cui, H.; Wu, Z. Energy consumption of cryptocurrency mining: A study of electricity consumption in mining cryptocurrencies. *Energy* **2019**, *168*, 160–168. [\[CrossRef\]](#)
38. Hajiaghapour-Moghim, M.; Hosseini, K.A.; Hajipour, E.; Vakilian, M. An approach to targeting cryptocurrency mining loads for energy efficiency enhancement. *IET Gener. Transm. Distrib.* **2022**, *16*, 4775–4790. [\[CrossRef\]](#)
39. Strezoski, L.; Stefani, I.; Brbaklic, B. Active management of distribution systems with high penetration of distributed energy resources. In Proceedings of the IEEE EUROCON 2019-18th International Conference on Smart Technologies, Novi Sad, Serbia, 1–4 July 2019; IEEE: New York, NY, USA, 2019.
40. Magdalena, R.; Mohammed, K.S.; Nassani, A.A.; Dascalu, N. Evaluating the environmental effects of bitcoin mining on energy and water use in the context of energy transition. *Sci. Rep.* **2025**, *15*, 8230. [\[CrossRef\]](#) [\[PubMed\]](#)
41. Bashari, M.; Doostkouei, S.G.; Fathabadi, M.; Soufimaajidpour, M. The environmental cost of cryptocurrency: Analyzing CO₂ emissions in the 9 leading mining countries. *Sustain. Futures* **2025**, *10*, 100792. [\[CrossRef\]](#)
42. Hutabarat, S. Bitcoin Mining and Renewable Energy: Navigating Sustainability, Profitability, and Electricity Market Dynamics. *Profitab. Electr. Mark. Dyn.* **2025**; ahead of print.
43. Asif, R.; Hassan, S.R. Shaping the future of Ethereum: Exploring energy consumption in Proof-of-Work and Proof-of-Stake consensus. *Front. Blockchain* **2023**, *6*, 1151724. [\[CrossRef\]](#)
44. Nasrinasrabadi, M.; Hejazi, M.A.; Chaharmahali, E.; Hussein, M. A comprehensive review of blockchain integration in smart grid with a special focus on internet of things. *Energy Convers. Manag. X* **2025**, *27*, 101196. [\[CrossRef\]](#)
45. Ali, F.; Khurram, M.U.; Sensoy, A.; Vo, X.V. Green cryptocurrencies and portfolio diversification in the era of greener paths. *Renew. Sustain. Energy Rev.* **2024**, *191*, 114137. [\[CrossRef\]](#)
46. Neeraj Shrivastava, P.M.B.M.; Lokeshwar Reddy, M.C.; Lavanya Banda, S.N.V.; Ramana, M.; Gayatri, P.; Siva Shankar, S. Optimizing Smart Grids for Distributed Energy Resource Integration and Management. *J. Inf. Syst. Eng. Manag.* **2025**, *10*, 421–427. [\[CrossRef\]](#)
47. Hosseinalizadeh, M.; Pordanjani, I.R.; Sayyadroushan, N.; Baneh, A.M.; Nasrinasrabadi, M.; Farbin, E. *Computing the Future: Research at the Convergence of Computer Engineering, Artificial Intelligence and Intelligent Technologies*; Nobel Sciences: Stockholm, Sweden, 2025.
48. Wongthongtham, P.; Marrable, D.; Abu-Salih, B.; Liu, X.; Morrison, G. Blockchain-enabled Peer-to-Peer energy trading. *Comput. Electr. Eng.* **2021**, *94*, 107299. [\[CrossRef\]](#)
49. Băra, A.; Oprea, S.-V.; Panait, M. Insights into Bitcoin and energy nexus. A Bitcoin price prediction in bull and bear markets using a complex meta model and SQL analytical functions. *Appl. Intell.* **2024**, *54*, 5996–6024. [\[CrossRef\]](#)
50. Okorie, D.I.; Gnatchiglo, J.M.; Weseh, P.K. Electricity and cryptocurrency mining: An empirical contribution. *Heliyon* **2024**, *10*, e33483. [\[CrossRef\]](#)
51. Bouri, E.; Saeed, T.; Vo, X.V.; Roubaud, D. Quantile connectedness in the cryptocurrency market. *J. Int. Financ. Mark. Inst. Money* **2021**, *71*, 101302. [\[CrossRef\]](#)
52. Harasheh, M.; Bouteska, A.; Hammad, M.H. Dynamic connectedness between energy markets and cryptocurrencies: Evidence from the COVID-19 pandemic. *J. Energy Mark.* **2023**, *16*, 1–27. [\[CrossRef\]](#)
53. Lu, Y.; Zheng, Z.; Tian, K.; Zhang, Y. A novel agent model of heterogeneous risk based on temporal interaction network for stock price simulation. *Phys. A Stat. Mech. Its Appl.* **2023**, *625*, 128981. [\[CrossRef\]](#)
54. Heydaritafreshi, O.; Moghim, M.H.; Hajipour, E. Utilization of Cryptocurrency Mining Load as High Flexible Load in Virtual Power Plant. *Electr. Power Syst. Res.* **2025**, *250*, 112074. [\[CrossRef\]](#)
55. Jain, A.; Gupta, S. Evaluation of electrical load demand forecasting using various machine learning algorithms. *Front. Energy Res.* **2024**, *12*, 1408119. [\[CrossRef\]](#)
56. Nakabi, T.A.; Toivanen, P. Deep reinforcement learning for energy management in a microgrid with flexible demand. *Sustain. Energy Grids Netw.* **2021**, *25*, 100413. [\[CrossRef\]](#)
57. Chen, H.; Xiong, R.; Lin, C.; Shen, W. Model predictive control based real-time energy management for hybrid energy storage system. *CSEE J. Power Energy Syst.* **2020**, *7*, 862–874. [\[CrossRef\]](#)
58. Parvathareddy, S.; Yahya, A.; Amuhaya, L.; Samikannu, R.; Suglo, R.S. A Hybrid Machine Learning and Optimization Framework for Energy Forecasting and Management. *Results Eng.* **2025**, *26*, 105425. [\[CrossRef\]](#)
59. Hosseina, M.; Moghaddam, M.S.; Hassannia, A. Optimizing energy and load management in island microgrids for enhancing resilience against resource interruptions. *Sci. Rep.* **2025**, *15*, 16297. [\[CrossRef\]](#)
60. SunPower. Available online: <https://us.sunpower.com/products/solar-panels> (accessed on 8 July 2025).
61. T.I.C. Ltd. Tumo-Int 10kW Wind Turbine. Available online: https://store.tumo-int.com/collections/wind-turbine-generator?srltid=AfmBOooSn5ddZhGC1IVLIT-SV-oFl5tLpUlaqK_u7MddVulocgO5bdi (accessed on 27 September 2025).

62. Caterpillar. C4.4 (50 Hz) | 50–110 kVA Diesel Generator. Available online: https://www.cat.com/en_US/products/new/power-systems/electric-power/diesel-generator-sets/1000001234.html (accessed on 27 September 2025).
63. U.S.E.I. Administration, Carbon Dioxide Emissions by Fuel. Available online: <https://www.eia.gov/environment/> (accessed on 27 September 2025).
64. Mårtensson, L. Emissions from Volvo's Trucks, Volvo Truck Corporation. Available online: https://www.volvotrucks.com/content/dam/volvo-trucks/markets/global/our-values/environmental-care/our-trucks/Emis_eng_10110_14001.pdf (accessed on 27 September 2025).
65. Pro, H. Advanced Kinetic Model. Available online: https://www.homerenergy.com/products/pro/docs/3.15/advanced_kinetic_model.html (accessed on 27 September 2025).
66. Pro, H. Kinetic Battery Model. Available online: https://www.homerenergy.com/products/pro/docs/3.15/kinetic_battery_model.html (accessed on 27 September 2025).
67. Pro, H. Storage Library. Available online: https://www.homerenergy.com/products/pro/docs/3.15/storage_library.html (accessed on 27 September 2025).
68. SMA Sunny Tripower 5.0 Smart Energy—Hybrid. Available online: <https://shop.natureenergieladen.de/SMA-Sunny-Tripower-5.0-Smart-Energy-Hybrid/BW-2021-06-002> (accessed on 27 September 2025).
69. ASICMinerValue. September 2024. Available online: <https://www.asicminervalue.com/> (accessed on 27 September 2025).
70. Rinaldi, F.; Moghaddampoor, F.; Najafi, B.; Marchesi, R. Economic feasibility analysis and optimization of hybrid renewable energy systems for rural electrification in Peru. *Clean. Technol. Environ. Policy* **2021**, *23*, 731–748. [CrossRef]
71. Colerick, K. Internal Rate of Return (IRR) Calculator. Available online: <https://www.calculatestuff.com/financial/irr-calculator> (accessed on 27 September 2025).
72. Küfeoğlu, S.; Özkuran, M. Bitcoin mining: A global review of energy and power demand. *Energy Res. Soc. Sci.* **2019**, *58*, 101273. [CrossRef]
73. Cengel, Y.A. *Thermodynamics: An Engineering Approach*; McGraw-Hill: Columbus, OH, USA, 2011.
74. Chandra, A.R.; Arora, R.C. *Refrigeration and Air Conditioning*; PHI Learning Pvt. Ltd.: Delhi, India, 2012.
75. Geng, H. *Data Center Handbook*; John Wiley & Sons: Hoboken, NJ, USA, 2014.
76. Louis, F.R.B.O.S. Federal Reserve Economic Data, Since 1991. Available online: <https://fred.stlouisfed.org/> (accessed on 27 September 2025).
77. Dunlop, J.P. *Photovoltaic Systems*; American Technical Publishers, Incorporated: Orland Park, IL, USA, 2012.
78. Duffie, J.A.; Beckman, W.A.; Worek, W. *Solar Engineering of Thermal Processes*; Wiley: Hoboken, NJ, USA, 1994.
79. Li, Z.; Huang, J.; Liaw, B.Y.; Zhang, J. On state-of-charge determination for lithium-ion batteries. *J. Power Sources* **2017**, *348*, 281–301. [CrossRef]
80. Jufri, F.H.; Aryani, D.R.; Garniwa, I.; Sudiarto, B. Optimal battery energy storage dispatch strategy for small-scale isolated hybrid renewable energy system with different load profile patterns. *Energies* **2021**, *14*, 3139. [CrossRef]
81. Puglia, G. Study and Development of a Web-Based Software for Hybrid Energy System Design and Solar Prediction Analysis. Ph.D. Dissertation, Università Politecnica delle Marche, Ancona, Italy, 2017. Available online: https://iris.univpm.it/retrieve/e18b8790-6cc8-d302-e053-1705fe0a27c8/tesi_puglia.pdf (accessed on 8 July 2025).
82. Lambert, T.; Gilman, P.; Lilienthal, P. Micropower system modeling with HOMER. *Integr. Altern. Sources Energy* **2006**, *1*, 379–385.
83. Bastian-Pinto, C.L.; Araujo, F.V.D.S.; Brandão, L.E.; Gomes, L.L. Hedging renewable energy investments with Bitcoin mining. *Renew. Sustain. Energy Rev.* **2021**, *138*, 110520. [CrossRef]
84. Asgari, N.; McDonald, M.T.; Pearce, J.M. Energy modeling and techno-economic feasibility analysis of greenhouses for tomato cultivation utilizing the waste heat of cryptocurrency miners. *Energies* **2023**, *16*, 1331. [CrossRef]
85. Alexander, C.K.; Sadiku, M.N. Electric circuits. *Transformation* **2000**, *135*, 4–5.
86. Edition, F.; Chapman, S.J. *Electric Machinery Fundamentals*; McGraw-Hill: Columbus, OH, USA, 2012.
87. Saadat, H. *Power System Analysis*; McGraw-Hill: Columbus, OH, USA, 1999; Volume 2.
88. Kreith, F.; Krumdieck, S. *Principles of Sustainable Energy Systems*; CRC Press: Boca Raton, FL, USA, 2013.
89. Rogers, G.F.C.; Mayhew, Y.R. *Engineering Thermodynamics, Work and Heat Transfer*; CRC Press: Boca Raton, FL, USA, 1992.
90. Everett, R.; Boyle, G.; Peake, S.; Ramage, J. *Energy Systems and Sustainability: Power for a Sustainable Future*; Oxford University Press: Oxford, UK, 2012.
91. Xin, Q. *Diesel Engine System Design*; Elsevier: Amsterdam, The Netherlands, 2011.
92. Gupta, H.N. *Fundamentals of Internal Combustion Engines*; PHI Learning Pvt. Ltd.: Delhi, India, 2012.
93. Li, Z.; Sun, H.; Xue, Y.; Li, Z.; Jin, X.; Wang, P. Resilience-Oriented Asynchronous Decentralized Restoration Considering Building and E-Bus Co-Response in Electricity-Transportation Networks. *IEEE Trans. Transp. Electrification* **2025**, *11*, 11701–11713. [CrossRef]

94. Du, Y.; Xue, Y.; Wu, W.; Shahidehpour, M.; Shen, X.; Wang, B.; Sun, H. Coordinated planning of integrated electric and heating system considering the optimal reconfiguration of district heating network. *IEEE Trans. Power Syst.* **2023**, *39*, 794–808. [[CrossRef](#)]
95. Lal, A.; You, F. Climate Sustainability through AI-Crypto Synergies and Energy Transition in the Digital Landscape to Cut 0.7 GtCO₂e by 2030. *Environ. Sci. Technol.* **2025**, *59*, 6061–6073. [[CrossRef](#)] [[PubMed](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.